

A Theory of Quantum Option Pricing and Empirical Tests

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Abstract

We develop a non-commutative quantum probability framework for option pricing when hedge trades move prices. Replacing the classical commutativity assumption with a canonical commutation relation—identified from the market’s price-impact slope—yields a closed-form pricing formula, resolves the short-maturity implied-volatility skew divergence, and supports a convex quantum state tomography procedure. Applied to regional bank equity options during a US banking crisis, the framework detects systemic contagion weeks before price-based indicators for gradually developing failures, and contemporaneously for sudden ones. A quantum entropy index Granger-causes implied volatility across multiple crisis cycles, complementing existing systemic risk measures.

Keywords: Quantum probability; option pricing; market microstructure; implied volatility skew; systemic risk.

JEL Classification: G13, G12, C61, G01, C65.

1 Introduction

The Black-Scholes-Merton paradigm (Black and Scholes, 1973; Merton, 1973) and its modern descendants—local volatility models (Dupire, 1994), stochastic volatility frameworks (Heston, 1993; Gatheral, 2006), and the rough volatility literature (Bayer et al., 2015; Gatheral et al., 2018)—have provided the conceptual foundation of derivatives pricing for half a century. Yet a structural assumption underlies all of these frameworks: the log-price process and the order-flow process are modeled within a separable Hilbert space in which the corresponding operators commute—that is, the joint evolution of prices and order flow can be represented by a commutative algebra of classical observables (Baaquie, 2004; Bouchaud and Potters, 2003; Ilinski, 2001; Föllmer and Sondermann, 1986). We argue that this commutativity assumption requires fundamental modification for markets with non-negligible microstructural frictions (Kyle, 1985; Almgren et al., 2005; Tóth et al., 2011; Glosten and Milgrom, 1985). Three structural anomalies motivate this reformulation: operator ordering ambiguity in delta-hedging strategies under finite price impact, the systematic failure of local and stochastic volatility models to reproduce the

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empirically observed $\mathcal{O}(\tau^{-1})$ short-maturity skew divergence, and the inability of conventional risk frameworks to capture latent systemic contagion within and across financial sectors during periods of market stress. The 2023 US Regional Banking Crisis, in which the Silicon Valley Bank, Signature Bank, and First Republic Bank failures produced cross-sectional patterns of implied volatility distortion that resisted standard calibration, exemplifies the limitations of the classical paradigm. These three anomalies are addressed in turn: Section 2 derives the non-commutative structure from price impact; Sections 3–4 resolve the skew divergence through the GNS pricing formula and semiclassical expansion; Section 5 addresses systemic contagion via quantum state tomography.

Our work draws upon and extends several strands of the mathematical finance and quantum probability literature. The first is the variance-optimal hedging tradition initiated by Föllmer and Sondermann (Föllmer and Sondermann, 1986) and developed by Schweizer (Schweizer, 1995, 2001), which formulates hedging in incomplete markets as a quadratic optimization problem in a Hilbert space of contingent claims. The Gelfand-Naimark-Segal (GNS) construction we develop generalizes this framework to the non-commutative setting, in which the inner product is induced by the market pricing functional rather than by an underlying probability measure. The second strand is the quantum finance literature, beginning with the path-integral formulation of Baaquie (Baaquie, 1997, 2004) and developed through the econophysics program of Bouchaud and collaborators (Bouchaud and Potters, 2003). While these contributions employ the mathematical apparatus of quantum mechanics as a computational device, our approach is conceptually distinct: we argue that the non-commutative structure of quantum probability is the natural algebraic framework for modeling markets with microstructural friction, and we derive the canonical commutation relation $[\hat{X}, \hat{P}] = i\hbar_{\text{econ}}\mathbf{1}$ from the economic properties of price impact rather than imposing it as a modeling assumption. The third strand is the rough volatility literature (Comte and Renault, 1998; Bayer et al., 2015; Gatheral et al., 2018; El Euch et al., 2018), which demonstrates that fractional Brownian motion with Hurst exponent $H < 1/2$ generates realistic short-maturity skew dynamics. Our framework predicts an $\mathcal{O}(\tau^{-1})$ short-maturity skew blow-up driven by $\hbar_{\text{econ}} > 0$, which provides a structural alternative to the $\mathcal{O}(\tau^{H-1/2})$ scaling of rough volatility models; the two mechanisms are parallel rather than nested. The fourth strand is the systemic risk measurement literature (Adrian and Brunnermeier, 2016; Brownlees and Engle, 2016; Acharya et al., 2017), which develops market-based indicators of financial fragility. Our quantum state tomography methodology contributes to this literature by providing a theoretically grounded construction of a forward-looking risk-regime barometer from a panel of macro-financial time series, with predictive horizons that extend beyond conventional volatility-based or correlation-based measures in our 2023 banking-crisis case study—a proof of concept whose out-of-sample generality remains to be established across additional crisis episodes.

We develop a non-commutative market framework in which the log-price observable $\hat{X}$ and the order-flow momentum observable $\hat{P}$ satisfy the canonical commutation relation

$$[\hat{X}, \hat{P}] = i\hbar_{\text{econ}}\mathbf{1}, \quad (1)$$

where the *economic Planck constant* $\hbar_{\text{econ}} > 0$ is a market-specific parameter capturing the irreducible scale of microstructural friction. We show that $\hbar_{\text{econ}}$ admits a direct microstructural interpretation as the product of Kyle’s price impact parameter λ and a reference order-flow unit $q_0 := \sigma_\varepsilon \sqrt{\Delta t}$, thereby connecting our quantum framework to the foundational market microstructure literature (Kyle, 1985). This paper makes three

principal contributions.

Theoretical Contribution: GNS Algebraic Pricing and Microstructural Uncertainty. We establish a rigorous non-commutative algebraic foundation for derivatives pricing under market microstructure frictions. By constructing the Gelfand-Naimark-Segal (GNS) triple $(\mathcal{H}_\omega, \pi_\omega, |\Omega\rangle)$ explicitly on the Weyl-Heisenberg algebra of market observables, we prove that arbitrage-free option valuation (under Assumption 3.1) admits the closed-form representation

$$P = \langle h^* | \Phi \rangle e^{-rT}, \quad (2)$$

where $|\Phi\rangle \in \mathcal{H}_\omega$ denotes the market’s quantum belief state in a Hilbert space, $|h^*\rangle$ denotes the optimal hedging state (using the lower-case notation of Sections 3–4), $\langle h^* | \Phi \rangle$ is the quantum transition amplitude, r is the risk-free rate, T is time to maturity, and the formula is evaluated at inception $t = 0$ (the general t -form is $e^{-r(T-t)}$). This formula constitutes the non-commutative analogue of the risk-neutral pricing rule, unifying hedging and valuation in a single Hilbert space structure, and reduces to the classical Black-Scholes price in the commutative limit $\hbar_{\text{econ}} \rightarrow 0$. The framework simultaneously formalizes the fundamental physical limits of market observation through a financial Heisenberg uncertainty relation $\sigma_X \sigma_P \geq \hbar_{\text{econ}}/2$, where $\hbar_{\text{econ}} > 0$ is the economic Planck constant that quantifies the irreducible scale of microstructural friction, generalizing classical pricing paradigms to markets with genuine microstructural barriers.

Methodological Contribution: Convex Quantum State Tomography. We develop a novel, computationally efficient empirical methodology—Quantum State Tomography (QST)—to reconstruct the market’s latent density operator ρ from observable asset moments. Instead of relying on heuristic parameter calibration, our QST procedure maps multi-asset macro-financial time series directly onto non-commuting Hermitian operators. The daily reconstruction of ρ is framed as the semidefinite program

$$\min_{\rho} \mathcal{D}(\rho, \hat{\rho}) \quad \text{s.t.} \quad \rho \succeq 0, \text{Tr}(\rho) = 1, \quad (3)$$

which optimizes a convex distance metric $\mathcal{D}$ on the cone of positive semidefinite matrices. We prove that this formulation exhibits computational complexity that scales polynomially with the number of observables, establishing a robust, non-parametric alternative to conventional structural calibration.

Empirical Contribution: Coherence Diagnostics and Semiclassical Skew Divergence. We conduct a comprehensive empirical validation by calibrating the framework to a panel of systemically important banks. First, we show that the off-diagonal elements of the reconstructed density matrix ρ , which capture quantum coherence between distressed and non-distressed financial sectors, act as a highly sensitive forward-looking early-warning indicator. For the SVB and Signature Bank failures (March 10 and 12), $\mathcal{C}(\rho_t)$ and the classical KRE–VIX Pearson correlation fire on the same day (March 9), reflecting the rapid onset of the March 2023 crisis. For the First Republic Bank failure (May 1)—the event with full HY-OAS credit-spread coverage and thus the cleanest test of the two-axis non-commutative measure— $\mathcal{C}(\rho_t)$ crosses its threshold on March 31, substantially before the classical Pearson correlation registers stress (May 1), providing a pre-emptive intervention window absent in classical, diagonal risk measures. Complementing this signal, the von Neumann entropy of the reconstructed density matrix

crosses its 95th-percentile baseline threshold approximately 15 trading days before the SVB failure, Granger-causing subsequent movements in KRE option-implied volatility and extending the actionable warning horizon further. Second, our framework predicts a short-maturity implied volatility skew exhibiting an $\mathcal{O}(\tau^{-1})$ divergence, which we derive via a semiclassical WKB expansion,

$$\sigma_{\text{eff}}^2(x, \tau) = \sigma_0^2 + \hbar_{\text{econ}} \gamma \frac{x}{\tau} + \hbar_{\text{econ}}^2 \beta \frac{1}{\tau^2} + \mathcal{O}(\hbar_{\text{econ}}^3), \quad (4)$$

where $x = \ln(S/K)$ is log-moneyness and $\tau = T - t$ is time-to-maturity. Here $\gamma := \omega([\hat{X}, \hat{P}]_+)$ and $\beta := \omega(\hat{X}^2)$ are state-dependent constants fixed by the GNS state ω ; the expansion is the semiclassical $\hbar_{\text{econ}} \rightarrow 0$ limit of the master formula (2), derived rigorously in Theorem 4.13 (eq. (25)) and Online Supplement §S.9 (the full smile-correction formula therein), resolving the short-term skew puzzle without invoking rough paths or jump processes. Finally, calibration on the SPDR S&P Regional Banking ETF (KRE) options-implied surface on March 10, 2023 demonstrates the model’s practical utility, with a marginal but theoretically consistent reduction in hedging errors over both Black-Scholes and Heston benchmarks in a single SVB stress-path case study, with the correction concentrated in out-of-the-money regimes where microstructural effects dominate.

The remainder of the paper is organized as follows. Section 2 develops the non-commutative algebraic foundation, including the microstructural derivation of the CCR, the Weyl-Heisenberg algebra, the GNS construction, and the financial uncertainty principle. Section 3 formulates the variance-optimal hedging problem as a quantum measurement and derives the main pricing formula. Section 4 establishes the duality between hedging and measurement, the semiclassical expansion, and the $\mathcal{O}(\tau^{-1})$ short-maturity skew divergence. Section 5 presents the empirical methodology, applies it to the 2023 US regional banking crisis, and reports robustness checks. Section 6 concludes with policy implications, limitations, and an outlook on future research. Technical proofs and supplementary derivations are collected in the appendices and Online Supplement.

Notation. Operators on the GNS Hilbert space $\mathcal{H}_\omega$ carry hats ($\hat{X}$, $\hat{P}$, $\hat{H}$). The symbol P denotes the option price (never a probability measure); H denotes the Hurst exponent only in comparisons with rough volatility; Δ is the hedge ratio (never a difference operator). A complete quantum-to-classical notation map is in Online Supplement Table S.0.

2 The Weyl–Heisenberg Algebra of Market Observables

2.1 Microstructural Origin of the Canonical Commutation Relation

Equation (1) is not imposed as a modeling assumption but *derived* from Kyle’s (1985) discrete-time price-impact model. Consider a single-period Kyle market with a risk-neutral market maker, an informed trader submitting order q_t , and noise traders. The market maker sets price as

$$X_{t+\Delta t} = X_t + \lambda q_t + \varepsilon_t, \quad (5)$$

where $\lambda > 0$ is the price-impact coefficient and ε_t is exogenous noise (Kyle, 1985). Define two operators on the trader's position state space $L^2(\mathbb{R})$:

$$\hat{X}_{\text{obs}} f(q) := (X_t + \lambda q) f(q), \quad \hat{P}_{\text{net}} f(q) := -i\sigma_\varepsilon \sqrt{\Delta t} \frac{\partial f}{\partial q}(q), \quad (6)$$

where $f \in L^2(\mathbb{R})$ is the trader's position wavefunction. $\hat{X}_{\text{obs}}$ represents the *post-trade* observable price $X_t + \lambda q$ (which depends on the order q through the price-impact rule (5)); $\hat{P}_{\text{net}}$ is the infinitesimal order-flow generator. The *observe-then-trade* sequence first applies $\hat{P}_{\text{net}}$ (shifting position by a unit order) then reads $\hat{X}_{\text{obs}}$ (recording the resulting price); the *trade-then-observe* sequence applies them in reverse order. The two operator compositions are

$$\begin{aligned} (\hat{X}_{\text{obs}} \hat{P}_{\text{net}}) f(q) &= (X_t + \lambda q) \left(-i\sigma_\varepsilon \sqrt{\Delta t} f'(q) \right), \\ (\hat{P}_{\text{net}} \hat{X}_{\text{obs}}) f(q) &= -i\sigma_\varepsilon \sqrt{\Delta t} \frac{\partial}{\partial q} [(X_t + \lambda q) f(q)] = -i\sigma_\varepsilon \sqrt{\Delta t} [(X_t + \lambda q) f'(q) + \lambda f(q)], \end{aligned} \quad (7)$$

where the second line uses the product rule and $\partial(\lambda q)/\partial q = \lambda$, reflecting the linear price-impact rule (5). Taking the difference yields

$$[\hat{X}_{\text{obs}}, \hat{P}_{\text{net}}] f(q) = (\hat{X}_{\text{obs}} \hat{P}_{\text{net}} - \hat{P}_{\text{net}} \hat{X}_{\text{obs}}) f(q) = i\lambda\sigma_\varepsilon \sqrt{\Delta t} f(q), \quad (8)$$

so that $[\hat{X}_{\text{obs}}, \hat{P}_{\text{net}}] = i\lambda\sigma_\varepsilon \sqrt{\Delta t} \mathbf{1} =: i\hbar_{\text{econ}} \mathbf{1}$, with $\hbar_{\text{econ}} := \lambda \cdot q_0$ and $q_0 := \sigma_\varepsilon \sqrt{\Delta t}$ a reference order-flow unit. This identifies $\hbar_{\text{econ}}$ as directly estimable from the slope of the empirical price-impact function (Almgren et al., 2005; Tóth et al., 2011). In Kyle's model, λ measures the permanent price impact per unit of order flow. Economically, $\hbar_{\text{econ}}$ quantifies the minimum distinguishable increment in price–order-flow space: below this scale, simultaneous observation of price and order flow is precluded by the trading mechanism itself, paralleling Heisenberg's measurement bound in physical systems. This connects the quantum friction scale directly to observable market microstructure depth.

The continuous-time limit $\Delta t \rightarrow 0$ with $\lambda \sim \Delta t^{-1/2}$ (consistent with the Almgren–Chriss scaling (Almgren et al., 2005)) yields the CCR (1). Formally, as $\Delta t \rightarrow 0$ with $\hbar_{\text{econ}} = \lambda\sigma_\varepsilon \sqrt{\Delta t}$ held fixed, the discrete operators $\hat{X}_{\text{obs}}^{(\Delta t)}$ and $\hat{P}_{\text{net}}^{(\Delta t)}$ converge in the strong resolvent sense to the canonical position and momentum operators on $L^2(\mathbb{R})$ satisfying $[\hat{X}, \hat{P}] = i\hbar_{\text{econ}} \mathbf{1}$ (Reed and Simon, 1972). The Trotter–Kato theorem (Kato, 1978) ensures that the semigroup generated by the limiting Lindblad equation is the *strong* operator limit (in the sense of pointwise convergence on each vector) of the discrete-time Markov semigroups, providing a rigorous foundation for the continuous-time framework. The frictionless limit $\lambda \rightarrow 0$ (or equivalently $\hbar_{\text{econ}} \rightarrow 0$) recovers a commutative algebra $[\hat{X}, \hat{P}] = 0$, which is precisely the classical BSM assumption (Black and Scholes, 1973; Merton, 1973). The CCR is therefore not a formal analogy from physics but a *derived consequence* of finite price impact in the sense of Kyle (Kyle, 1985) and Glosten–Milgrom (Glosten and Milgrom, 1985).

Proposition 2.1 (Non-commutativity is necessary under price impact). *Suppose the price and order-flow operators $\hat{X}$ and $\hat{P}$ satisfy Kyle's linear price-impact rule (5) with $\lambda > 0$. Then:*

- (i) $[\hat{X}, \hat{P}] = i\hbar_{\text{econ}} \mathbf{1} \neq 0$ (non-commutativity is forced by the market mechanism).
- (ii) If one instead imposes $[\hat{X}, \hat{P}] = 0$ (commutative algebra), then the operator $\hat{P}$ cannot generate bounded translations of $\hat{X}$ in $L^2(\mathbb{R})$, and no self-financing strategy

$\delta_t \in \mathcal{K}$ satisfies the Kyle pricing rule for all order sizes $q \in \mathbb{R}$. Equivalently, perfect dynamic replication is unattainable for any payoff with non-trivial order-flow sensitivity.

Proof. Part (i) is established in Section 2.1. For part (ii): if $[\hat{X}, \hat{P}] = 0$, the Stone–von Neumann theorem (Proposition 2.15) does not apply and the algebra $\mathcal{A}$ is commutative, i.e., isomorphic to $C_0(\mathbb{R}^2)$ (continuous functions vanishing at infinity on phase space). In a commutative algebra, every element is a function of commuting variables, so $\hat{P}$ cannot generate non-trivial phase-space rotations. Concretely, the composition identity $\hat{X}_{\text{obs}}\hat{P}_{\text{net}}f = \hat{P}_{\text{net}}\hat{X}_{\text{obs}}f$ would require $\lambda = 0$ (from (8)), contradicting $\lambda > 0$. Hence, under Kyle price impact, the commutative framework is internally inconsistent. $\square$

Remark 2.2 (Why not a stochastic control formulation?). An alternative response to price impact is to augment the classical Itô SDE with a feedback term $dX_t = \sigma dW_t + \lambda \delta_t dt$, where δ_t is the hedge ratio. This approach—used in Almgren–Chriss (2001) and related execution-cost models—treats price impact as an exogenous cost function and preserves the commutative probability space. The key distinction is that such models assume the hedge ratio δ_t and the post-trade price $X_t + \lambda \delta_t$ can be simultaneously observed: δ_t and X_t commute as random variables. Proposition 2.1 shows this simultaneous observability fails when price impact is linear in order flow: observing the post-trade price already incorporates the order, and the order in turn responds to the pre-trade price, generating irreducible non-commutativity at the level of the algebra of observables rather than at the level of stochastic paths.

Remark 2.3 (Continuous-time scaling and Back’s equilibrium). The scaling $\lambda \sim \Delta t^{-1/2}$ used above follows the Almgren–Chriss execution-cost framework (Almgren et al., 2005), in which market impact is an *exogenous* cost function. In the continuous-time generalisation of Kyle’s equilibrium (Kyle, 1985), the endogenous price-impact coefficient satisfies $\lambda_t = \sigma_v / (2\sigma_u \sqrt{T-t})$, which has a different time-dependence. The present paper uses the Almgren–Chriss scaling to fix $\hbar_{\text{econ}}$ as a time-invariant parameter; this is a modelling choice that simplifies the empirical calibration at the cost of abstracting from the time-decay of λ_t in Back’s equilibrium. The calibrated value $\hbar_{\text{econ}} = 0.087$ should therefore be interpreted as a *time-averaged* friction parameter over the calibration window, rather than an instantaneous equilibrium coefficient.

Remark 2.4 (Interpretation). Equation (8) shows that the post-trade price and the magnitude of the executed order (operators defined in (6)) cannot be simultaneously resolved with arbitrary precision in any market with $\lambda > 0$. This is the financial analogue of the Heisenberg uncertainty principle: measuring X_t precisely requires a transaction that perturbs P_t , and vice versa. The economic Planck constant $\hbar_{\text{econ}}$ quantifies the irreducible cost of microstructural friction, and vanishes only in the frictionless limit $\lambda \rightarrow 0$.

Definition 2.5 (Economic Planck constant). Let $\hat{X}$ be the log-price operator and $\hat{P}$ be the order-flow momentum operator, defined such that $\hat{P}$ generates translations in $\hat{X}$ via the Weyl commutation relation (Weyl, 1928). The *economic Planck constant* is defined as

$$\hbar_{\text{econ}} := \lambda \cdot q_0 = \lambda \cdot \sigma_\varepsilon \sqrt{\Delta t}, \quad (9)$$

where $\lambda > 0$ is Kyle’s price-impact coefficient, σ_ε is the standard deviation of order-flow noise, and Δt is the minimum trading time scale. By construction, $\hbar_{\text{econ}}$ satisfies

$$[\hat{X}, \hat{P}] = i\hbar_{\text{econ}}\mathbf{1}, \quad (10)$$

where $\mathbf{1}$ is the identity operator on the market Hilbert space, as derived in Section 2.1.

The commutation relation (10) is the mathematical expression of the fact that price and order flow are complementary observables in a market with microstructure frictions; $\hbar_{\text{econ}}$ can be estimated directly from the slope of the empirical price impact function (Remark 2.6).

Remark 2.6 (Units and interpretation). In dimensional terms, $\hbar_{\text{econ}} = \lambda \cdot q_0$ carries the units of price $\times$ order-flow, i.e., [price] $\times$ [shares]. In the log-return / normalized-order-flow parametrization used throughout this paper—where prices are expressed as log-returns and order flow is normalized by $\sigma_\varepsilon \sqrt{\Delta t}$ —both observables become dimensionless, so $\hbar_{\text{econ}}$ is effectively dimensionless in this convention. Typical values estimated from market data range from 0.01 to 0.15, with higher values corresponding to more illiquid markets.

Remark 2.7 (Notation). We use $\hbar_{\text{econ}}$ as the canonical symbol throughout.

Theorem 2.8 (Financial uncertainty principle). *Let $\hat{X}$ be the log-price operator and $\hat{P}$ be the order-flow momentum operator, satisfying the canonical commutation relation $[\hat{X}, \hat{P}] = i\hbar_{\text{econ}}\mathbf{1}$. Then for any market state ω , the variances of $\hat{X}$ and $\hat{P}$ satisfy*

$$\sigma_X \cdot \sigma_P \geq \frac{\hbar_{\text{econ}}}{2},$$

where $\sigma_X^2 = \omega(\hat{X}^2) - \omega(\hat{X})^2$ and $\sigma_P^2 = \omega(\hat{P}^2) - \omega(\hat{P})^2$.

Proof. Since $\hat{X}$ and $\hat{P}$ are unbounded operators, the Robertson inequality applies on the dense domain $\mathcal{D} := \mathcal{D}(\hat{X}\hat{P}) \cap \mathcal{D}(\hat{P}\hat{X}) \subset \mathcal{H}_\omega$, which is non-empty and dense in the Schrödinger representation by the Stone–von Neumann theorem (von Neumann, 1931). For any $|\psi\rangle \in \mathcal{D}$, the Cauchy–Schwarz inequality applied to the commutator $[\hat{X}, \hat{P}]$ in the GNS representation gives

$$\|(\hat{X} - \langle \hat{X} \rangle)|\psi\rangle\| \cdot \|(\hat{P} - \langle \hat{P} \rangle)|\psi\rangle\| \geq \frac{1}{2} |\langle \psi | [\hat{X}, \hat{P}] | \psi \rangle| = \frac{\hbar_{\text{econ}}}{2}.$$

Taking the infimum over all $|\psi\rangle \in \mathcal{D}$ with $\| |\psi\rangle \| = 1$ and identifying σ_X^2, σ_P^2 with the respective variances in state ω yields the inequality for all states ω whose GNS vector lies in $\mathcal{D}$. $\square$

2.2 The Weyl Form of the Canonical Commutation Relation

Equation (10) is unbounded and presents domain issues in infinite dimensions. The Weyl form exponentiates the unbounded operators to unitary operators, yielding a rigorous algebraic formulation.

Definition 2.9 (Weyl operators). For each $(a, b) \in \mathbb{R}^2$, define the Weyl operator

$$W(a, b) = \exp\left(\frac{i}{\hbar_{\text{econ}}}(a\hat{X} + b\hat{P})\right). \quad (11)$$

Here the first parameter a multiplies $\hat{X}$ (log-price) and the second b multiplies $\hat{P}$ (order-flow momentum). Under this convention and $[\hat{X}, \hat{P}] = i\hbar_{\text{econ}}\mathbf{1}$, the Baker–Campbell–Hausdorff formula yields the commutator $[a\hat{X} + b\hat{P}, a'\hat{X} + b'\hat{P}] = i\hbar_{\text{econ}}(ab' - a'b)\mathbf{1}$, giving the Weyl relation below.

The Weyl operators (11) satisfy the Weyl form of the CCR:

$$W(a, b)W(a', b') = \exp\left(+\frac{i}{2\hbar_{\text{econ}}}(ab' - a'b)\right) W(a + a', b + b'). \quad (12)$$

The Weyl operators generate the Weyl–Heisenberg algebra $\mathcal{A}$, which is the C^* -algebra of market observables. The algebraic structure captures the non-commutative nature of price and order flow: two Weyl operators commute if and only if the corresponding phase-space area $ab' - a'b$ is an integer multiple of $2\pi\hbar_{\text{econ}}$.

Definition 2.10 (Market observable algebra). The *market observable algebra* $\mathcal{A}$ is the C^* -algebra generated by the Weyl operators $\{W(a, b) : (a, b) \in \mathbb{R}^2\}$, equipped with the involution $W(a, b)^* = W(-a, -b)$ and the unique C^* -norm determined by the CCR (12).

2.3 The GNS Construction for Market States

A market state is a positive, normalized, linear functional ω on $\mathcal{A}$ that assigns to each observable its expected value. The Gelfand–Naimark–Segal (GNS) construction represents this state as a vector in a Hilbert space.

Proposition 2.11 (GNS construction for market states (Takesaki, 2003)). *Let ω be a faithful normal state on the market observable algebra $\mathcal{A}$. Then there exists a triple $(\mathcal{H}_\omega, \pi_\omega, |\Omega_\omega\rangle)$ consisting of:*

- (i) a Hilbert space $\mathcal{H}_\omega$;
- (ii) a $*$ -representation $\pi_\omega : \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H}_\omega)$;
- (iii) a cyclic vector $|\Omega_\omega\rangle \in \mathcal{H}_\omega$ such that $\overline{\pi_\omega(\mathcal{A})|\Omega_\omega\rangle} = \mathcal{H}_\omega$ and $\omega(A) = \langle \Omega_\omega | \pi_\omega(A) | \Omega_\omega \rangle$ for all $A \in \mathcal{A}$.

This triple is unique up to unitary equivalence. The GNS construction is a standard result of C^ -algebra theory; the inner product on $\mathcal{H}_\omega$ is defined by $\langle \pi_\omega(A)\Omega_\omega | \pi_\omega(B)\Omega_\omega \rangle = \omega(A^*B)$, and the representation is extended by continuity.*

The GNS Hilbert space $\mathcal{H}_\omega$ is the space of all financial positions that can be expressed as limits of portfolios of market observables. The cyclic vector $|\Omega_\omega\rangle$ represents the *vacuum state* of the market —the ground state of the financial system in the absence of priced risk factors.

2.4 Hedging Portfolios as Quantum Observables

We now formulate the hedging problem within the GNS framework. Let $G(S_T)$ be the payoff of a European option maturing at T , and let $Y = e^{-rT}G(S_T)$ be its discounted value. Via the GNS construction, the discounted payoff is represented as a vector $|Y\rangle \in \mathcal{H}_\omega$ defined by

$$|Y\rangle = \pi_\omega(e^{-rT}\hat{G})|\Omega_\omega\rangle,$$

where $\hat{G} \in \mathcal{A}$ is the observable corresponding to the payoff function.

Definition 2.12 (Hedgeable subspace). Let $\mathcal{K} \subset \mathcal{H}_\omega$ be the closed linear subspace spanned by the discounted asset prices available for hedging:

$$\mathcal{K} = \overline{\text{span}}\{|S_{t_i}\rangle : t_i \in [0, T]\},$$

where $|S_{t_i}\rangle$ is the GNS vector corresponding to the discounted asset price at time t_i . We call $\mathcal{K}$ the *hedgeable subspace*.

A self-financing hedging portfolio is represented by an operator $H \in \mathcal{A}$ whose GNS image $\pi_\omega(H)$ acts on $\mathcal{H}_\omega$. The optimal hedge is the orthogonal projection of the payoff vector onto the hedgeable subspace.

Theorem 2.13 (Quantum pricing formula). *Let $G(S_T)$ be the payoff of a European option maturing at T , let $r \geq 0$ be the risk-free rate, and let $|\Omega_\omega\rangle \in \mathcal{H}_\omega$ be the cyclic vector of the GNS representation. Then the arbitrage-free price of the option at time $t \leq T$ is given by the quantum expectation value*

$$P_t = \langle \Omega_\omega | \pi_\omega(H) | \Omega_\omega \rangle e^{-r(T-t)}, \quad (13)$$

where $H \in \mathcal{A}$ is the self-adjoint operator representing the optimal self-financing hedging portfolio, and $\pi_\omega(H)$ is its GNS representation. Equivalently, in the State-Hedge representation (Corollary 4.8), the optimal hedge representative $h^* = \Pi_{\mathcal{K}_t} Y_t$ is the orthogonal projection of the payoff-state vector $Y_t = g\Phi_t$ onto the hedgeable subspace $\mathcal{K}_t$, and the price is the discounted real inner product $P_t = \langle h^* | \Phi_t \rangle e^{-r(T-t)}$ on the real Hilbert space $\mathcal{H}_\omega \simeq L^2(\mathbb{R}, \mathbb{R})$.

Proof. The proof proceeds in three steps.

Step 1: GNS representation of the payoff. The discounted payoff $Y = e^{-rT}G(S_T)$ is represented as a vector $|Y\rangle \in \mathcal{H}_\omega$ via

$$|Y\rangle = \pi_\omega(e^{-rT}\hat{G})|\Omega_\omega\rangle.$$

Step 2: Optimal hedge as orthogonal projection. The optimal hedge $|h^*\rangle$ is the unique vector in $\mathcal{K}$ that minimizes the squared hedging error $\|Y - h\|_\omega^2$. By the Hilbert space projection theorem, this is the orthogonal projection of $|Y\rangle$ onto $\mathcal{K}$:

$$|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle,$$

where $\Pi_{\mathcal{K}} \in \mathcal{B}(\mathcal{H}_\omega)$ is the orthogonal projection operator. Since $|\Omega_\omega\rangle$ is cyclic for $\pi_\omega(\mathcal{A})$, the orbit $\pi_\omega(\mathcal{A})|\Omega_\omega\rangle$ is dense in $\mathcal{H}_\omega$. As $\mathcal{K}$ is a closed subspace and the projection $\Pi_{\mathcal{K}}$ is continuous, there exists a unique self-adjoint operator $H \in \mathcal{A}$ (in the closure of the GNS image) such that $\pi_\omega(H)|\Omega_\omega\rangle = |h^*\rangle$.

Step 3: Discounting and no-arbitrage. The time- t value of the hedge is the discounted expectation:

$$P_t = \langle \Omega_\omega | \pi_\omega(H) | \Omega_\omega \rangle e^{-r(T-t)}.$$

The market pricing functional $\omega_t(A) = \langle \Omega_\omega | \pi_\omega(A) | \Omega_\omega \rangle$ is positive and normalised by the GNS construction. Its faithfulness—and hence the equivalence with the absence of arbitrage—is established in Theorem 3.11 (Section 3), proved independently of this formula. $\square$

Remark 2.14 (Relation to the classical BSM formula). In the classical limit $\hbar_{\text{econ}} \rightarrow 0$, the canonical commutation relation (12) becomes $[\hat{X}, \hat{P}] = 0$, the observable algebra $\mathcal{A}$ is commutative, and the GNS Hilbert space reduces to $L^2(\mathbb{P})$. The hedgeable subspace $\mathcal{K}$ becomes the full space, so $\Pi_{\mathcal{K}} = \mathbf{1}$ and the pricing formula collapses to

$$P_t^{\text{BSM}} = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[G(S_T) | \mathcal{F}_t],$$

where $\mathbb{Q}$ is the unique risk-neutral measure. The quantum correction is therefore a genuine generalization that contains BSM as the commutative special case.

2.5 The Market Hamiltonian and Lindblad Dynamics

The dynamics of market observables are generated by a self-adjoint Hamiltonian operator $\hat{H}$. In the Heisenberg picture, the time evolution of an observable $A \in \mathcal{A}$ is given by

$$\frac{d}{dt}A(t) = \frac{i}{\hbar_{\text{econ}}}[\hat{H}, A(t)] + \mathcal{L}[A(t)],$$

where $\mathcal{L}$ is a Lindbladian superoperator that accounts for non-unitary, dissipative dynamics (Lindblad, 1976; Gorini et al., 1976). The Lindbladian takes the standard form

$$\mathcal{L}[A] = \sum_k \gamma_k \left(L_k^\dagger A L_k - \frac{1}{2} \{L_k^\dagger L_k, A\} \right),$$

where $\{L_k\}$ are Lindblad operators representing the sources of market friction and $\gamma_k \geq 0$ are relaxation rates.

For the purpose of option pricing, we consider the Hamiltonian

$$\hat{H} = \frac{\hat{P}^2}{2M} + V(\hat{X}), \quad (14)$$

where $M > 0$ is the *market depth* parameter and $V(\hat{X})$ is a potential function representing the confining effect of arbitrage bounds. The kinetic term $\hat{P}^2/(2M)$ captures the cost of order flow: larger order flow generates larger price impact, with the market depth M determining the scale of this effect. The potential $V(\hat{X})$ ensures that the log-price remains within economically meaningful bounds.

The state of the market at time t is described by a density operator ρ_t acting on $\mathcal{H}_\omega$. In the Schrödinger picture, the evolution of the state is governed by the Lindblad master equation:

$$\frac{d}{dt}\rho_t = -\frac{i}{\hbar_{\text{econ}}}[\hat{H}, \rho_t] + \sum_k \gamma_k \left(L_k \rho_t L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho_t\} \right). \quad (15)$$

The market risk state $|\Phi_t\rangle$ that appears in the pricing formula is obtained from the density operator via a purification: there exists a Hilbert space $\mathcal{H}_\omega \otimes \mathcal{H}_{\text{anc}}$ such that

$$\rho_t = \text{Tr}_{\text{anc}}(|\Phi_t\rangle\langle\Phi_t|),$$

where Tr_{anc} denotes the partial trace over an ancillary space representing unobserved market degrees of freedom. The purified state $|\Phi_t\rangle$ evolves under a stochastic Schrödinger equation driven by the Lindblad dynamics, and the pricing formula (13) generalizes to

$$P_t = \langle\Phi_t|\pi_\omega(H) \otimes \mathbf{1}_{\text{anc}}|\Phi_t\rangle e^{-r(T-t)},$$

which is the duality pairing that forms the computational basis for the empirical methodology developed in Section 5.

2.6 Uniqueness of the Quantum Representation (Stone–von Neumann)

A natural question is whether the adoption of the quantum Hilbert-space formalism is a genuine consequence of the CCR, or merely one of many possible mathematical representations. The Stone–von Neumann theorem resolves this: it shows that the quantum mechanical Hilbert-space framework is the *unique* irreducible representation of the Weyl CCR algebra up to unitary equivalence.

Proposition 2.15 (Uniqueness of the Weyl representation). *Let $\mathcal{A}$ be the Weyl C^* -algebra*

generated by $\{W(a, b) : (a, b) \in \mathbb{R}^2\}$ satisfying (12) with $\hbar_{\text{econ}} > 0$. Then every strongly continuous irreducible unitary representation $(\mathcal{H}, W)$ of $\mathcal{A}$ is unitarily equivalent to the Schrödinger representation on $L^2(\mathbb{R}, \mathbb{C})$:

$$(W(a, b)f)(x) = e^{i(ax + ab\hbar_{\text{econ}}/2)/\hbar_{\text{econ}}} f(x + b), \quad f \in L^2(\mathbb{R}, \mathbb{C}).$$

In particular, the Hilbert-space structure $\mathcal{H}_\omega \simeq L^2(\mathbb{R}, \mathbb{C})$ is not an arbitrary modelling choice but the unique irreducible carrier space of the CCR derived in Section 2.1.

Proof. This is the classical Stone–von Neumann theorem (von Neumann, 1931; Reed and Simon, 1972). The key hypotheses—strong continuity of $a \mapsto W(a, 0)$ and $b \mapsto W(0, b)$ (Weyl regularity) and irreducibility—are satisfied in the financial setting: strong continuity follows from the continuity of the price-impact rule (5) in the order size q , and irreducibility reflects the fact that a single informed trader interacting with the full price-impact mechanism generates the entire orbit of the observable algebra. The Schrödinger representation is therefore the canonical and unique framework for quantising the market observables derived from Kyle’s model. $\square$

Remark 2.16 (Implications for modelling). Proposition 2.15 addresses the concern that the quantum probability framework is merely a formal rewriting of classical operator algebra. Any other representation of $\mathcal{A}$ with $\hbar_{\text{econ}} > 0$ decomposes as a direct sum (or direct integral) of copies of the Schrödinger representation; the irreducible component is always the quantum Hilbert space. The commutative ($\hbar_{\text{econ}} = 0$) limit is excluded by Proposition 2.15, as the Stone–von Neumann theorem does not apply when the CCR degenerates to $[\hat{X}, \hat{P}] = 0$ —consistent with the fact that the frictionless market admits a classical ($L^2(\Omega, P)$) rather than quantum representation.

3 No-Arbitrage and the Quantum Pricing Formula

The Weyl-Heisenberg algebra of Section 2 provides the algebraic foundation for a rigorous formulation of the hedging problem under price impact. We now show that minimizing quadratic hedging error in this non-commutative setting is equivalent to an orthogonal projection in the GNS Hilbert space, yielding a closed-form quantum pricing formula that reduces to the classical Black–Scholes rule in the commutative limit $\hbar_{\text{econ}} \rightarrow 0$.

Assumption 3.1 (Market state in the hedgeable subspace). The market risk state Φ_t lies in the hedgeable subspace $\mathcal{K}_t$: $\Phi_t \in \mathcal{K}_t$. This regularity condition ensures that the market state itself can be expressed as a limit of tradeable portfolio positions. It is formally justified and verified for the single-qubit parametrisation in Section 4 (Assumption 4.7).

3.1 The Hedging Problem with Price Impact

In frictionless BSM markets, any contingent claim can be perfectly replicated by a self-financing portfolio. This relies on the assumption that hedging trades do not affect prices. In real markets, every hedge trade alters the underlying price, creating a feedback loop that we formalize through the non-commutative algebra of market observables.

Let $\mathcal{A}$ be the C^* -algebra generated by the log-price operator $\hat{X}$ and the order-flow momentum operator $\hat{P}$, satisfying the Weyl form of the canonical commutation relation (12). The market state is described by a faithful normal state ω on $\mathcal{A}$, which we identify with the GNS cyclic vector $|\Omega_\omega\rangle$ in the representation $\pi_\omega : \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H}_\omega)$.

Definition 3.2 (Hedgeable subspace, restated). Let $\mathcal{K} \subset \mathcal{H}_\omega$ be the closed linear subspace spanned by the discounted asset prices available for hedging:

$$\mathcal{K} = \overline{\text{span}}\{|S_{t_i}\rangle : t_i \in [0, T]\},$$

where $|S_{t_i}\rangle$ is the GNS vector corresponding to the discounted asset price at time t_i . We call $\mathcal{K}$ the *hedgeable subspace*.

Lemma 3.3 (Non-triviality of the hedgeable subspace). *Under the Stone–von Neumann representation $\mathcal{H}_\omega \simeq L^2(\mathbb{R}, \mathbb{C})$ of Proposition 2.15, and assuming the discounted asset price process $(S_t)_{t \in [0, T]}$ is non-degenerate (i.e., S_t is not almost-surely constant), the hedgeable subspace $\mathcal{K}$ satisfies $\{0\} \subsetneq \mathcal{K} \subsetneq \mathcal{H}_\omega$.*

Proof. *Non-emptiness* ($\mathcal{K} \neq \{0\}$): Each $|S_{t_i}\rangle$ is the GNS image of the non-zero observable $\hat{S}_{t_i} \in \mathcal{A}$; faithfulness of ω gives $\omega(\hat{S}_{t_i}^* \hat{S}_{t_i}) > 0$, so $\| |S_{t_i}\rangle \|^2 = \omega(\hat{S}_{t_i}^* \hat{S}_{t_i}) > 0$.

Properness ($\mathcal{K} \neq \mathcal{H}_\omega$): The wavefunction of $|S_{t_i}\rangle$ in the log-price representation is localised near the asset level S_{t_i} (it belongs to a compact subset of the spectrum of $\hat{X}$). The payoff $g(x) = (e^x - K)_+$ of a call option grows without bound as $x \rightarrow +\infty$, so $|g\rangle = \pi_\omega(\hat{G})|\Omega_\omega\rangle \notin \mathcal{K}$ for generic $K > 0$. Hence $\mathcal{K}$ is a proper subspace, confirming market incompleteness for $\hbar_{\text{econ}} > 0$. $\square$

The hedging problem is to find a portfolio $|h\rangle \in \mathcal{K}$ that minimizes the quadratic hedging error:

$$\min_{|h\rangle \in \mathcal{K}} \| |Y\rangle - |h\rangle \|^2_\omega, \quad (16)$$

where $|Y\rangle = e^{-rT}|G(S_T)\rangle$ is the discounted payoff vector.

Theorem 3.4 (Existence and uniqueness of the optimal hedge). *The hedging problem (16) has a unique solution $|h^*\rangle \in \mathcal{K}$, given by the orthogonal projection of $|Y\rangle$ onto $\mathcal{K}$:*

$$|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle, \quad (17)$$

where $\Pi_{\mathcal{K}} \in \mathcal{B}(\mathcal{H}_\omega)$ is the orthogonal projection operator onto $\mathcal{K}$.

Proof. The functional $J(h) = \|Y - h\|_\omega^2$ is strictly convex and coercive on the closed subspace $\mathcal{K}$. By the Hilbert space projection theorem, there exists a unique minimizer $|h^*\rangle \in \mathcal{K}$ satisfying the orthogonality condition

$$\langle Y - h^* | k \rangle_\omega = 0 \quad \forall |k\rangle \in \mathcal{K}.$$

This condition is equivalent to $|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle$. $\square$

Corollary 3.5 (Minimum hedging error). *The minimum achievable hedging error is*

$$\varepsilon_{\min}^2 = \|Y\|_\omega^2 - \|\Pi_{\mathcal{K}}Y\|_\omega^2. \quad (18)$$

In the classical frictionless limit $\hbar_{\text{econ}} \rightarrow 0$, $\mathcal{K}$ becomes the full space $\mathcal{H}_\omega$ and $\varepsilon_{\min} = 0$, recovering BSM perfect replication.

3.2 Hedging as Quantum Measurement

Equation (16) is equivalent to a quantum measurement: the hedging portfolio projects the payoff onto the subspace of hedgeable claims (von Neumann, 1932).

Definition 3.6 (Hedging dual vector). Let $|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle$ be the optimal hedge vector. The *hedging dual vector* $\langle h^*|$ is the unique element of the dual space $\mathcal{H}_\omega^*$ obtained via the Riesz representation theorem: for any $|\psi\rangle \in \mathcal{H}_\omega$, $\langle h^*|\psi\rangle = \langle h^*|\psi\rangle_\omega$.

Proposition 3.7 (Hedging as a quantum measurement). *The optimal hedge $|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle$ is the orthogonal projection of the payoff-state vector onto the hedgeable subspace:*

$$|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle = \Pi_{\mathcal{K}}\pi_{\omega}(\hat{Y})|\Omega_{\omega}\rangle,$$

where $\Pi_{\mathcal{K}} : \mathcal{H}_{\omega} \rightarrow \mathcal{K}$ is the orthogonal projection operator onto the hedgeable subspace and $|Y\rangle = \pi_{\omega}(\hat{Y})|\Omega_{\omega}\rangle$ is the GNS representative of the discounted payoff. At the operator level, the optimal hedge satisfies $\hat{h}^* = \mathbb{E}_{\omega}[\hat{Y} | \mathcal{A}_{\mathcal{K}}]$ where $\mathbb{E}_{\omega}[\cdot | \mathcal{A}_{\mathcal{K}}]$ is Takesaki's conditional expectation onto the sub-algebra $\mathcal{A}_{\mathcal{K}}$ associated with $\mathcal{K}$ (Takesaki, 2003); at the GNS vector level this is equivalent to $|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle$ as above.

Proof (deferred to Theorem 4.2 in §4). The result follows as a corollary of the Hedging–Measurement Duality (Theorem 4.2 in Section 4), which is proved independently of this proposition. That theorem establishes $\pi_{\omega}(\hat{h}^*)|\Omega_{\omega}\rangle = \Pi_{\mathcal{K}}\pi_{\omega}(\hat{Y})|\Omega_{\omega}\rangle$ in the GNS representation, from which taking the inner product with $|\Omega_{\omega}\rangle$ (and using $\omega(A) = \langle\Omega_{\omega}|\pi_{\omega}(A)|\Omega_{\omega}\rangle$) yields the trace formula for $|h^*\rangle$. $\square$

3.3 The Main Pricing Formula

Theorem 3.8 (Quantum pricing formula). *Let $G(S_T)$ be the payoff of a European option maturing at T , let $r \geq 0$ be the risk-free rate, and let $|\Phi_t\rangle \in \mathcal{H}_{\omega} \otimes \mathcal{H}_{\text{anc}}$ be the purified market risk state at time $t \leq T$, evolving under the Lindblad master equation (15). Then the arbitrage-free price of the option at time t is*

$$P_t = \langle h^* | \Phi_t \rangle e^{-r(T-t)}, \quad (19)$$

where:

- (i) $|Y\rangle = e^{-rT}|G(S_T)\rangle$ is the discounted payoff vector;
- (ii) $|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle$ is the optimal hedge vector, the orthogonal projection of $|Y\rangle$ onto $\mathcal{K}$;
- (iii) $\langle h^* |$ is the Riesz dual vector of $|h^*\rangle$;
- (iv) $e^{-r(T-t)}$ is the deterministic discount factor.

Corollary 4.8 specializes this formula to the real Hilbert space $\mathcal{H}_{\omega} \simeq L^2(\mathbb{R}, \mathbb{R})$, making realness and positivity manifest and providing the operational form used for empirical calibration.

Proof. The proof proceeds in five steps.

Step 1: GNS representation of the payoff. By the GNS construction, the discounted payoff $Y = e^{-rT}G(S_T)$ is represented as $|Y\rangle = \pi_{\omega}(e^{-rT}\hat{G})|\Omega_{\omega}\rangle$.

Step 2: Optimal hedge as orthogonal projection. By Theorem 3.4, $|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle$.

Step 3: Duality pairing and the market risk state. The market risk state $|\Phi_t\rangle$ purifies ρ_t : $\rho_t = \text{Tr}_{\text{anc}}(|\Phi_t\rangle\langle\Phi_t|)$. The value of the optimal hedge at time t is $V_t^{\text{hedge}} = \text{Tr}(\rho_t \hat{h}^*) = \langle\Phi_t|\hat{h}^* \otimes \mathbf{1}_{\text{anc}}|\Phi_t\rangle = \langle h^* | \Phi_t \rangle$.

Step 4: Discounting. $P_t = e^{-r(T-t)}V_t^{\text{hedge}}$.

Step 5: No-arbitrage consistency. The market pricing functional $\omega_t(A) = \langle \Phi_t | \pi_\omega(A) | \Phi_t \rangle$ is positive, normalized, and faithful. By Theorem 3.11, this is equivalent to the absence of arbitrage. $\square$

Remark 3.9 (Relation to the BSM formula). In the classical limit $\hbar_{\text{econ}} \rightarrow 0$, $[\hat{X}, \hat{P}] = 0$, the GNS Hilbert space reduces to $L^2(\mathbb{P})$, $\mathcal{K} = \mathcal{H}_\omega$, $\Pi_{\mathcal{K}} = \mathbf{1}$, and $|h^*\rangle = |Y\rangle$. The pricing formula collapses to $P_t^{\text{BSM}} = e^{-r(T-t)} \mathbb{E}^\mathbb{Q}[G(S_T) | \mathcal{F}_t]$, recovering the classical BSM price.

3.4 The No-Arbitrage Condition in the Quantum Framework

The fundamental theorem of quantum asset pricing extends the classical results of Harrison and Kreps (1979) and Delbaen and Schachermayer (1994) to the non-commutative setting: it states that a faithful normal state on the observable algebra is equivalent to the absence of quantum arbitrage.

Definition 3.10 (Quantum arbitrage). A *quantum arbitrage* is a non-zero observable $A \in \mathcal{A}$ such that $A \geq 0$ and $\omega(A) = 0$ for some market state ω . In other words, an arbitrage is a non-negative financial position that has zero cost but is not identically zero.

Theorem 3.11 (Fundamental theorem of quantum asset pricing). *Under Assumption 3.1 ($\Phi_t \in \mathcal{K}_t$), the following are equivalent:*

- (i) *There exists a faithful normal state ω on $\mathcal{A}$.*
- (ii) *The market is free of quantum arbitrage.*
- (iii) *The GNS representation π_ω is faithful and the pricing functional $P_t = \langle h^* | \Phi_t \rangle e^{-r(T-t)}$ is strictly positive for all non-zero non-negative payoffs.*

Proof. We prove the equivalence in three steps.

(i) $\Rightarrow$ (ii). Suppose ω is faithful and $A \geq 0$ with $\omega(A) = 0$. By faithfulness, $A = 0$, so no non-zero arbitrage exists.

(ii) $\Rightarrow$ (i). Assume the market is free of quantum arbitrage (condition (ii)). By Online Supplement §S.13 (the NFLVR–GNS equivalence theorem therein), the NFLVR condition—of which no-quantum-arbitrage is the quantum analogue—implies that the market state ω is a faithful normal state on $\mathcal{A}$. Specifically, the argument in Online Supplement §S.13 constructs ω via Gleason’s theorem on the Type I von Neumann algebra $\pi_\omega(\mathcal{A})'' = \mathcal{B}(\mathcal{H}_\omega)$: complete additivity of the valuation functional on projections (ensured by no-arbitrage) yields a unique trace-class density operator ρ with $\omega = \text{Tr}(\rho \cdot)$ and $\rho > 0$ (faithfulness). This establishes (i).

Remark 3.12 (Gleason’s theorem and the two-dimensional case). Gleason’s theorem (Gleason, 1957) requires $\dim(\mathcal{H}) \geq 3$. For the single-qubit parametrisation ($\dim = 2$) used in the empirical Section 5, the Bloch-sphere representation provides a direct alternative: every state on $\mathcal{B}(\mathbb{C}^2)$ has the form $\omega(A) = \frac{1}{2} \text{Tr}((I + \mathbf{r} \cdot \boldsymbol{\sigma})A)$ with $\|\mathbf{r}\| \leq 1$, and faithfulness is equivalent to $\|\mathbf{r}\| < 1$. The absence of arbitrage in the two-dimensional case is verified directly: $\omega(A) = 0$ for $A \geq 0$ implies $\text{Tr}(\rho A) = 0$, which with $\rho > 0$ forces $A = 0$. Theorem 3.11 therefore holds for all $\dim(\mathcal{H}) \geq 2$, with the Gleason argument applying when $\dim \geq 3$ and the explicit Bloch construction covering $\dim = 2$.

(i) $\Leftrightarrow$ (iii). If ω is faithful and normal, the GNS representation $\pi_\omega : \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H}_\omega)$ is injective (faithful). Strict positivity of P_t follows because for any non-zero non-negative payoff $\hat{G} \geq 0$, faithfulness gives $\omega(\hat{G}) > 0$, and by Assumption 3.1 ($\Phi_t \in \mathcal{K}_t$) together with the positivity argument of Remark 4.10, $P_t = \langle h^* | \Phi_t \rangle e^{-r(T-t)} = \langle Y_t | \Phi_t \rangle e^{-r(T-t)} > 0$ (since $\varepsilon_t \perp \mathcal{K}_t$ by construction and $\Phi_t \in \mathcal{K}_t$ by Assumption 3.1, so $\varepsilon_t \perp \Phi_t$). Conversely, if $P_t > 0$ for all non-zero non-negative payoffs, then $\omega(\hat{G}) > 0$ for all such $\hat{G}$, which is precisely faithfulness of ω . $\square$

Corollary 3.13 (No-arbitrage and the economic Planck constant). *For $\hbar_{\text{econ}} > 0$, the Weyl algebra $\mathcal{A}$ is simple: it has no non-trivial closed two-sided ideals. Consequently, every non-zero $*$ -representation of $\mathcal{A}$ is faithful (injective). In particular, the GNS representation π_ω for any non-zero state ω is faithful, so the pricing formula is well-defined.*

Remark 3.14. Simplicity of $\mathcal{A}$ implies that every representation is faithful, but not that every state is faithful in the sense that $\omega(A^*A) = 0 \Rightarrow A = 0$. A vector state $\omega_\psi(A) = \langle \psi | A | \psi \rangle$ satisfies faithfulness only if ψ is a separating vector (i.e. $A\psi = 0 \Rightarrow A = 0$). For the Schrödinger representation on $L^2(\mathbb{R})$, any non-zero ψ is separating for bounded operators, so vector states are faithful in practice. The faithful normal state whose existence is guaranteed by Theorem 3.11(i) is a generic element of the state space that satisfies this separating property by construction.

In the classical limit $\hbar_{\text{econ}} \rightarrow 0$, the algebra becomes commutative and faithfulness of ω reduces to the classical condition that the pricing measure be equivalent to the physical measure.

Proof. The simplicity of the Weyl C^* -algebra for $\hbar_{\text{econ}} \neq 0$ is a standard result (Petz, 1990). Injectivity of every non-zero representation follows directly from simplicity: if $\ker(\pi) \neq \{0\}$, then $\ker(\pi)$ is a non-trivial closed two-sided ideal, contradicting simplicity. $\square$

3.5 The Gibbs State and the KMS Condition

In thermal equilibrium, quantum systems are described by Gibbs states. The financial analogue is a market in which the pricing functional is determined by a Hamiltonian $\hat{H}$ and an inverse temperature $\beta = 1/(k_B T)$, where k_B is the Boltzmann constant and T is the “market temperature.”

Definition 3.15 (Gibbs market state). A *Gibbs market state* at inverse temperature $\beta > 0$ is a state ω_β on $\mathcal{A}$ defined by

$$\omega_\beta(A) = \frac{\text{Tr}(e^{-\beta \hat{H}} A)}{\text{Tr}(e^{-\beta \hat{H}})}, \quad A \in \mathcal{A},$$

where $\hat{H}$ is the market Hamiltonian (14).

The Gibbs state satisfies the Kubo–Martin–Schwinger (KMS) condition, which characterizes thermal equilibrium in quantum statistical mechanics (Kubo, 1957; Martin and Schwinger, 1959). In the financial context, the KMS condition implies that the market is in a state of “no asymptotic arbitrage” in the sense that no trading strategy can generate risk-free profits over infinitely long horizons.

Theorem 3.16 (KMS condition and no asymptotic arbitrage). *Let ω_β be a Gibbs market state at inverse temperature β . Then for any $A, B \in \mathcal{A}$, there exists a function $F_{A,B}(z)$*

analytic in the strip $0 < \Im(z) < \beta$ and continuous on its closure such that

$$F_{A,B}(t) = \omega_\beta(A\alpha_t(B)), \quad F_{A,B}(t + i\beta) = \omega_\beta(\alpha_t(B)A),$$

where $\alpha_t(A) = e^{it\hat{H}} A e^{-it\hat{H}}$ is the Heisenberg time evolution. This condition is equivalent to the absence of asymptotic arbitrage.

Proof. The KMS condition is a standard result for Gibbs states: for $\omega_\beta(A) = Z^{-1} \text{Tr}(e^{-\beta\hat{H}} A)$, the function $F_{A,B}(z) = Z^{-1} \text{Tr}(e^{-\beta\hat{H}} A e^{iz\hat{H}} B e^{-iz\hat{H}})$ is analytic in the strip $\{0 < \Im(z) < \beta\}$ and satisfies the boundary conditions by the cyclicity of the trace (Bratteli and Robinson, 1981).

For the equivalence with no asymptotic arbitrage: an asymptotic arbitrage is a sequence of strategies $\{H_n\} \subset \mathcal{A}$ with $\omega_\beta(H_n^* H_n) \rightarrow 0$ (vanishing cost) but $\liminf_{n \rightarrow \infty} \omega_\beta((H_n - \varepsilon)_+) > 0$ for some $\varepsilon > 0$ (bounded profit). If such a sequence existed, then for large n we have $H_n \approx 0$ in the $L^2(\omega_\beta)$ sense, so $F_{H_n,B}(z) = \omega_\beta(H_n \alpha_z(B)) \rightarrow 0$ uniformly on compacts by the Cauchy estimate for bounded analytic functions. But the boundary value $F_{H_n,B}(t + i\beta) = \omega_\beta(\alpha_t(B) H_n) \rightarrow \omega_\beta(\alpha_t(B) \cdot 0) = 0$ as well. Since $\omega_\beta(H_n^* H_n) \rightarrow 0$ implies $H_n \rightarrow 0$ in the $L^2(\omega_\beta)$ -norm (weak operator topology), the continuous functional calculus is weakly continuous (Bratteli and Robinson, 1981): $f(H_n) \rightarrow f(0)$ weakly for every continuous f . Applying $f(x) = (x - \varepsilon)_+$ gives $(H_n - \varepsilon)_+ \rightarrow 0$ weakly, hence $\omega_\beta((H_n - \varepsilon)_+) \rightarrow 0$, contradicting $\liminf_{n \rightarrow \infty} \omega_\beta((H_n - \varepsilon)_+) > 0$. Therefore no asymptotic arbitrage can exist for any Gibbs state ω_β . $\square$

Remark 3.17 (Economic interpretation of the KMS condition). The KMS condition has a natural economic interpretation: it states that the market is in a state of “thermal equilibrium” where the pricing functional is time-translation invariant. In this equilibrium, the price of an option is determined by the Hamiltonian $\hat{H}$ and the inverse temperature β , which together encode the market’s risk preferences and the cost of microstructural friction. The KMS condition also provides a rigorous foundation for the use of the Lindblad master equation (15) in modeling the evolution of the market state: the Gibbs state is the stationary solution of the Lindblad dynamics in the absence of external driving.

3.6 The Pricing Formula in the Gibbs State

When the market is in a Gibbs state, the pricing formula takes a particularly elegant form. The optimal hedge vector $|h^*\rangle$ can be expressed directly in terms of the Hamiltonian and the inverse temperature.

Proposition 3.18 (Gibbs state pricing). *Let ω_β be a Gibbs market state at inverse temperature β , defined by $\omega_\beta(A) = Z^{-1} \text{Tr}(e^{-\beta\hat{H}} A)$ where $\hat{H}$ is the market Hamiltonian and $Z = \text{Tr}(e^{-\beta\hat{H}})$. Then the optimal hedge vector in the pricing formula (19) is*

$$|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle,$$

where $|Y\rangle$ is the discounted payoff vector. This formulation guarantees that the inner product $\langle h^* | \Phi_t \rangle$ is manifestly real and non-negative for any non-negative payoff.

Proof. For a KMS state ω_β with respect to the time-evolution automorphism group $\alpha_t(A) = e^{it\hat{H}} A e^{-it\hat{H}}$, the Tomita–Takesaki modular operator satisfies $\Delta = e^{-\beta\hat{H}}$ (Bratteli and Robinson, 1981), i.e., the modular operator coincides with the Gibbs weight. In the GNS representation of ω_β , the cyclic vector $|\Omega_\beta\rangle$ is a separating vector for $\pi_{\omega_\beta}(\mathcal{A})$, and the modular conjugation J satisfies $J\pi_\omega(A)J = \pi_\omega(A^*)^*$ on $\mathcal{H}_{\omega_\beta}$. The optimal hedge

$|h^*\rangle = \Pi_{\mathcal{K}}|Y\rangle$ follows from Theorem 4.2; the Gibbs structure of ω_β ensures that the KMS analyticity condition holds (see also Theorem 3.16), which guarantees both the faithfulness of ω_β (by Theorem 3.11) and the strict positivity of the pricing formula.

Remark 3.19. The equation $\pi_\omega(e^{-\beta\hat{H}})|\Omega_\omega\rangle = |h^*\rangle$ holds specifically when the payoff operator $\hat{G}$ commutes with the modular flow, i.e., when $\hat{G}$ is in the centraliser of ω_β . For general payoffs, $|h^*\rangle$ is obtained by projection as above.

□

3.7 Dynamics of the Pricing Formula

The pricing formula (19) evolves in time as the market risk state $|\Phi_t\rangle$ evolves under the Lindblad master equation. This evolution provides a dynamic hedging strategy that is consistent with the no-arbitrage condition.

Proposition 3.20 (Dynamics of the option price). *The option price $P_t = \langle h^*|\Phi_t\rangle e^{-r(T-t)}$ evolves according to*

$$dP_t = rP_t dt + \langle h^*|d\Phi_t\rangle e^{-r(T-t)},$$

where $d\Phi_t$ is the stochastic differential of the market risk state under the Lindblad dynamics (15). The hedge ratio is given by the duality pairing $\Delta_t = \langle h^*|\partial_S\Phi_t\rangle e^{-r(T-t)}$.

Proof. In this proposition h^* is treated as the time- t reference hedge, fixed at the projection $h^* = \Pi_{\mathcal{K}_t}Y_t$ (i.e., recalibrated at each t but treated as constant for the infinitesimal increment). Differentiating $P_t = \langle h^*|\Phi_t\rangle e^{-r(T-t)}$ with respect to t yields the discount term $rP_t dt$ plus the stochastic increment $\langle h^*|d\Phi_t\rangle e^{-r(T-t)}$, where $d\Phi_t$ is governed by the Itô form of the Lindblad master equation. The hedge ratio $\Delta_t = \langle h^*|\partial_S\Phi_t\rangle e^{-r(T-t)}$ follows from the chain rule applied to the S -dependence of Φ_t . The full derivation of the Itô–Lindblad stochastic differential and the delta formula is given in Online Supplement S.1.2. □

4 Duality Hedging and the Semiclassical Expansion

Section 3 established the *existence* and uniqueness of the optimal hedge as an orthogonal projection in the GNS Hilbert space. This section provides its *explicit form* and derives the empirically testable consequences. Sections 4.1–4.2 show that the projection is a quantum measurement, giving the State–Hedge pricing representation. Section 4.3 then expands the pricing formula in powers of $\hbar_{\text{econ}}$, producing the $\mathcal{O}(\tau^{-1})$ short-maturity skew divergence and the quantum optimal hedge—the two theoretical predictions that drive the empirical tests of Section 5.

4.1 The Duality Between Hedging and Measurement

The central duality of the quantum framework equates the optimal hedge for a European option to a quantum measurement of the payoff operator in the GNS representation, yielding a computationally tractable pricing approach.

Definition 4.1 (Hedging operator). Let $\hat{G} = g(\hat{S}_T)$ be the payoff operator for a European option with payoff function g and let $\rho \in \mathcal{T}(\mathcal{H})$ be the market state. The *optimal hedge* is the operator $\hat{\Pi} \in \mathcal{A}$ that minimizes the quadratic hedging error

$$\hat{\Pi}^* = \arg \min_{\hat{\Pi} \in \mathcal{A}} \text{Tr}[\rho(\hat{G} - \hat{\Pi})^2], \quad (20)$$

where $\mathcal{A}$ is the observable algebra generated by $\{\hat{X}, \hat{P}\}$.

The optimization (20) is a least-squares problem in the non-commutative $L^2(\rho)$ space (Pisier, 2003). The solution is the conditional expectation of $\hat{G}$ onto $\mathcal{A}$ with respect to ρ , which we denote by $\mathbb{E}_\rho[\hat{G} | \mathcal{A}]$.

Theorem 4.2 (Hedging–measurement duality). *Let $\rho \in \mathcal{T}(\mathcal{H})$ be a faithful normal state on $\mathcal{A}$ and let $\pi_\rho : \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H}_\rho)$ be the GNS representation. Then for any payoff operator $\hat{G} \in \mathcal{A}$, the optimal hedge $\hat{\Pi}^*$ satisfies*

$$\pi_\rho(\hat{\Pi}^*)|\Omega_\rho\rangle = P_{\mathcal{A}_\rho}\pi_\rho(\hat{G})|\Omega_\rho\rangle, \quad (21)$$

where $P_{\mathcal{A}_\rho}$ is the orthogonal projection onto the closure of $\pi_\rho(\mathcal{A})|\Omega_\rho\rangle$ in $\mathcal{H}_\rho$, and $|\Omega_\rho\rangle$ is the GNS cyclic vector.

Proof. The GNS representation provides an isometric embedding of $\mathcal{A}$ into $\mathcal{B}(\mathcal{H}_\rho)$ via π_ρ , with cyclic vector $|\Omega_\rho\rangle$ such that $\omega(A) = \langle \Omega_\rho | \pi_\rho(A) \Omega_\rho \rangle$ for all $A \in \mathcal{A}$. The quadratic hedging error (20) can be written as

$$\text{Tr}[\rho(\hat{G} - \hat{\Pi})^2] = \|\pi_\rho(\hat{G})|\Omega_\rho\rangle - \pi_\rho(\hat{\Pi})|\Omega_\rho\rangle\|_{\mathcal{H}_\rho}^2.$$

The minimizer is the orthogonal projection of $\pi_\rho(\hat{G})|\Omega_\rho\rangle$ onto the closed subspace $\overline{\pi_\rho(\mathcal{A})|\Omega_\rho\rangle}$, which is precisely $P_{\mathcal{A}_\rho}\pi_\rho(\hat{G})|\Omega_\rho\rangle$. Since π_ρ is injective (faithfulness of ρ), there exists a unique $\hat{\Pi}^* \in \mathcal{A}$ satisfying (21). $\square$

Remark 4.3 (Economic interpretation). The duality theorem reveals that the optimal hedge is the *quantum measurement* of the payoff operator in the market state ρ . The GNS vector $|\Omega_\rho\rangle$ represents the “vacuum” state of the market, and the projection $P_{\mathcal{A}_\rho}$ extracts the component of the payoff that is replicable by trading in the available securities. The non-replicable component, orthogonal to $\mathcal{A}_\rho$, is the intrinsic hedging error that cannot be eliminated by any dynamic trading strategy.

4.2 The State–Hedge Pricing Representation

The core pricing formula is formalized as a representation theorem on a real Hilbert space $\mathcal{H}_\omega \simeq L^2(\mathbb{R}, \mathbb{R})$ over the log-price axis $x = \log S_T$, guaranteeing real-valued inner products and a manifestly positive pricing functional.

Definition 4.4 (Payoff-state vector). Let $g(x)$ be the payoff function of a European option expressed in terms of the log-price $x = \log S_T$, and let $\Phi_t(x)$ be the wavefunction of the market risk state at time t in the log-price representation. The *payoff-state vector* at time t is the pointwise product

$$Y_t(x) := g(x) \Phi_t(x) \in L^2(\mathbb{R}, \mathbb{R}), \quad (22)$$

where the Hilbert space $\mathcal{H}_\omega \simeq L^2(\mathbb{R}, \mathbb{R})$ is a *real* L^2 space over the log-price axis, equipped with the standard real inner product

$$\langle f | h \rangle = \int_{\mathbb{R}} f(x) h(x) dx.$$

The real nature of $\mathcal{H}_\omega$ guarantees that all duality pairings are strictly real-valued.

Definition 4.5 (Hedgeable subspace at time t). Let $\mathcal{K}_t \subset \mathcal{H}_\omega$ be the closed linear subspace spanned by the discounted asset-price wavefunctions available for hedging at time t :

$$\mathcal{K}_t = \overline{\text{span}}\{\psi_{S_{t_i}}(x) : t_i \in [0, t]\},$$

where $\psi_{S_{t_i}}(x)$ is the wavefunction of the discounted asset price at time t_i . The subspace $\mathcal{K}_t$ represents the set of all tradeable hedging strategies.

Lemma 4.6 (Closedness of hedgeable subspace and market-state regularity). *Under the Stone–von Neumann representation $\mathcal{H}_\omega \simeq L^2(\mathbb{R}, \mathbb{R})$, $\mathcal{K}_t$ as defined above is a proper closed subspace of $\mathcal{H}_\omega$ whenever the market depth is finite ($\hbar_{\text{econ}} > 0$). Moreover, the Lindblad-evolved market risk state satisfies $\Phi_t \in \mathcal{K}_t$ under the following regularity assumption.*

Assumption 4.7 (Market state in hedgeable subspace: formal statement). The market risk state Φ_t lies in the hedgeable subspace: $\Phi_t \in \mathcal{K}_t$ (cf. Assumption 3.1 in Section 3). This holds whenever the Lindblad dynamics preserve the asset-price sub-algebra (i.e., the market state is generated by traded observables), as is the case for the single-qubit Bloch parametrization of Section 5.

Proof of Lemma 4.6. Closedness. By definition, $\mathcal{K}_t$ is the closed linear span of the set $\{\psi_{S_{t_i}} : t_i \in [0, t]\}$, hence closed by construction.

Properness. We claim $\mathcal{K}_t \subsetneq \mathcal{H}_\omega = L^2(\mathbb{R}, \mathbb{R})$ whenever the market is genuinely incomplete ($\hbar_{\text{econ}} > 0$). Each wavefunction $\psi_{S_{t_i}}(x)$ is concentrated near the asset-price level S_{t_i} : it belongs to $L^2(\mathbb{R})$ but its support is localized in price-space. The payoff function $g(x)$ of a generic option has support that extends beyond the span of these wavefunctions (e.g., for a call, $g(x) = (e^x - K)_+$ grows at $+\infty$, while any finite linear combination of localized wavefunctions decays). Therefore $g \notin \mathcal{K}_t$ in general, confirming $\mathcal{K}_t \subsetneq L^2(\mathbb{R})$.

Remark on incompleteness. Note that the irreducibility of the Weyl algebra representation (Stone–von Neumann theorem) refers to the absence of *operator-invariant* closed subspaces, not the absence of arbitrary closed subspaces. $L^2(\mathbb{R})$ has infinitely many proper closed subspaces (e.g., $\{f : \text{supp}(f) \subseteq [0, 1]\}$), and $\mathcal{K}_t$ is one of them. Irreducibility is not invoked here. $\square$

Corollary 4.8 (State–Hedge Pricing Representation, of Theorem 3.8). *Let $g(x)$ be the payoff of a European option maturing at T , let $r \geq 0$ be the risk-free rate, and let $\Phi_t \in \mathcal{H}_\omega$ be the market risk state at time $t \leq T$, evolving under the Lindblad master equation (15). Define the payoff-state vector $Y_t = g\Phi_t$ as in (22), and let $h^* \in \mathcal{K}_t$ be the unique optimal hedge representative given by the orthogonal projection of Y_t onto the hedgeable subspace:*

$$h^* := \Pi_{\mathcal{K}_t} Y_t \in \mathcal{K}_t, \tag{23}$$

where $\Pi_{\mathcal{K}_t} : \mathcal{H}_\omega \rightarrow \mathcal{K}_t$ is the orthogonal projection operator. Then by Theorem 3.8, the arbitrage-free price is given by formula (19), where $\langle \cdot | \cdot \rangle$ is interpreted as the real inner product on $\mathcal{H}_\omega \simeq L^2(\mathbb{R}, \mathbb{R})$, a real Hilbert space over the log-price axis $x = \log S_T$.

Remark 4.9 (Real versus complex Hilbert space). The abstract GNS construction produces a *complex* Hilbert space $\mathcal{H}_\omega^\mathbb{C}$, as required by the spectral theory of the Weyl operators and the Robertson inequality. The present corollary works with its *realification* $\mathcal{H}_\omega := \mathcal{H}_\omega^\mathbb{C}|_\mathbb{R}$, the same underlying vector space equipped with the real inner product $\langle f, g \rangle_\mathbb{R} := \text{Re}\langle f, g \rangle_\mathbb{C}$. This is legitimate because option payoffs $g(x)$ and market risk states $\Phi_t(x)$ in the log-price representation are *real-valued* functions, so the complex inner product reduces to a real integral. The realification preserves the Hilbert-space projection

theorem (hence the existence and uniqueness of h^*) and guarantees $P_t = \langle h^* | \Phi_t \rangle_{\mathbb{R}} \in \mathbb{R}_{\geq 0}$ without loss of the operator-algebraic structure of §2. The Stone–von Neumann theorem (von Neumann, 1931) ensures that the irreducible representation of the CCR algebra on $L^2(\mathbb{R}, \mathbb{C})$ is unique up to unitary equivalence; the real subspace used here is the invariant subspace of real-valued functions, which is preserved by the option pricing projection.

Proof. By Theorem 3.8, the arbitrage-free price is $P_t = \langle h^* | \Phi_t \rangle e^{-r(T-t)}$ on the GNS Hilbert space $\mathcal{H}_\omega$. The GNS construction (Definition 2.11) realizes $\mathcal{H}_\omega$ concretely as the real $L^2(\mathbb{R}, \mathbb{R})$ space over the log-price axis $x = \log S_T$, with $\Phi_t(x)$ the wavefunction representative of the market risk state and $Y_t(x) = g(x)\Phi_t(x)$ (Definition 4.4) the payoff-state vector. The orthogonal projection $h^* = \Pi_{\mathcal{K}_t}(e^{-rT}Y_t)$ onto the hedgeable subspace $\mathcal{K}_t$ (Definition 4.5) and the commutativity of the deterministic discount factor with the projection yield the pricing formula (19). Realness and positivity follow immediately from the real Hilbert space structure. $\square$

Remark 4.10 (Economic interpretation of the State–Hedge representation). The State–Hedge Pricing Representation (19) provides a complete resolution of the three core economic requirements:

Realness. Because $\mathcal{H}_\omega \simeq L^2(\mathbb{R}, \mathbb{R})$ is a real Hilbert space, the inner product $\langle h^* | \Phi_t \rangle$ is manifestly real-valued. No complex phases appear in the pricing formula; the complex structure of the underlying C^* -algebra is confined to the algebraic relations among observables and does not affect the final price.

Positivity. The optimal hedge $h^* = \Pi_{\mathcal{K}_t}Y_t$ is the orthogonal projection of the payoff-state vector $Y_t = g\Phi_t$ onto $\mathcal{K}_t$. Write the decomposition

$$Y_t = h^* + \varepsilon_t, \quad \varepsilon_t \perp \mathcal{K}_t,$$

where ε_t is the unhedgeable residual. Since $\Phi_t \in \mathcal{K}_t$ (the market risk state is itself a tradeable quantity), the orthogonality $\varepsilon_t \perp \Phi_t$ implies

$$\langle h^* | \Phi_t \rangle = \langle Y_t | \Phi_t \rangle = \int_{\mathbb{R}} g(x) |\Phi_t(x)|^2 dx \geq 0,$$

where the inequality follows from $g(x) \geq 0$ for a standard option payoff and $|\Phi_t(x)|^2 \geq 0$. Thus the price is non-negative for any non-negative payoff, establishing the positivity of the pricing functional.

Classical limit (complete markets). When the market is frictionless ($\hbar_{\text{econ}} = 0$), the hedgeable subspace coincides with the full space: $\mathcal{K}_t = \mathcal{H}_\omega$. In this case $\Pi_{\mathcal{K}_t} = \mathbf{1}$, so $h^* = Y_t = g\Phi_t$, and the pricing formula collapses to

$$P_t^{\text{BSM}} = e^{-r(T-t)} \langle g\Phi_t | \Phi_t \rangle = e^{-r(T-t)} \int_{\mathbb{R}} g(x) |\Phi_t(x)|^2 dx = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[g(S_T) | \mathcal{F}_t],$$

where $|\Phi_t(x)|^2 = q_t(x)$ is the classical risk-neutral density. The quantum formula therefore contains the classical BSM expectation as the special case of complete markets.

Incomplete markets (general case). When $\hbar_{\text{econ}} > 0$, the market is incomplete and $\mathcal{K}_t$ is a proper subspace of $\mathcal{H}_\omega$. The optimal hedge h^* is the best tradeable approximation to the payoff-state vector Y_t , and the inner product $\langle h^* | \Phi_t \rangle$ represents the valuation of this optimal hedge under the market friction encoded by $\hbar_{\text{econ}}$. The residual $\varepsilon_t = Y_t - h^*$

is the intrinsic hedging error that cannot be eliminated by any dynamic strategy—it is the cost of non-commutativity, quantified by the economic Planck constant.

Corollary 4.11 (Relation to the GNS pricing formula). *The State–Hedge representation (19) is equivalent to the abstract pricing relation (2) realized in the GNS Hilbert space, with $|h^*\rangle = \Pi_K|Y\rangle$, where $|Y\rangle$ is the GNS vector of the discounted payoff. The real L^2 representation makes the realness and positivity properties manifest.*

4.3 The Semiclassical Expansion and the $\mathcal{O}(\tau^{-1})$ Skew

A semiclassical (WKB) expansion in $\hbar_{\text{econ}}$ yields tractable closed-form corrections to the implied volatility surface. The detailed derivation is given in Online Supplement §S.1; we state the main result here.

Assumption 4.12 (Semiclassical regime). The economic Planck constant satisfies $\hbar_{\text{econ}} \ll 1$, and the market state ρ is a perturbation of a classical state. Specifically, we assume that the Wigner function $W_\rho(x, p)$ of the market state admits an expansion

$$W_\rho(x, p) = W_{\text{cl}}(x, p) + \hbar_{\text{econ}} W_1(x, p) + \hbar_{\text{econ}}^2 W_2(x, p) + \mathcal{O}(\hbar_{\text{econ}}^3), \quad (24)$$

where $W_{\text{cl}}(x, p)$ is a classical probability density on the phase space $(x, p) \in \mathbb{R}^2$.

Under this assumption, the expectation of any observable $A \in \mathcal{A}$ can be expanded as

$$\omega(A) = \int_{\mathbb{R}^2} a_{\text{cl}}(x, p) W_{\text{cl}}(x, p) dx dp + \mathcal{O}(\hbar_{\text{econ}}),$$

where $a_{\text{cl}}(x, p)$ is the classical symbol of A . This is the non-commutative analogue of the method of characteristics for classical stochastic processes.

Theorem 4.13 (Effective implied volatility). *Under Assumption 4.12, the implied volatility σ_{imp} of a European option with log-moneyness $x = \log(S_0/K)$ (consistent with §1, eq. (4)) and time-to-maturity τ admits the short-maturity expansion*

$$\sigma_{\text{imp}}^2(x, \tau) = \sigma_0^2 + \hbar_{\text{econ}} \cdot \gamma \cdot \frac{x}{\tau} + \hbar_{\text{econ}}^2 \cdot \beta \cdot \frac{1}{\tau^2} + \mathcal{O}(\hbar_{\text{econ}}^3), \quad (25)$$

where $\sigma_0^2 = \omega(\hat{P}^2)$ is the classical variance, $\gamma = \omega([\hat{X}, \hat{P}]_+)$ is the skew coefficient, and $\beta = \omega(\hat{X}^2)$ is the convexity coefficient.

Remark 4.14 (Skew sign convention and comparison with classical models). The skew coefficient $\gamma = \omega([\hat{X}, \hat{P}]_+) > 0$ in the equity leverage-effect regime. The quantum model produces an $\mathcal{O}(\tau^{-1})$ short-maturity skew, faster than BSM (no skew), Heston ($\mathcal{O}(1)$ finite), and rough volatility ($\mathcal{O}(\tau^{H-1/2})$ for $H < 1/2$). Full proofs, the model-comparison table, and the Bloch sphere worked example are in Online Supplement S.1. Empirical calibration is in Section 5.

Proof. The proof proceeds via a Wigner–Moyal phase-space expansion in $\hbar_{\text{econ}}$ followed by term-by-term identification of the implied-volatility corrections. The detailed three-step calculation is given in Online Supplement S.1.1. $\square$

Corollary 4.15 (Skew explosion). *For any $\hbar_{\text{econ}} > 0$,*

$$\lim_{\tau \rightarrow 0} \frac{\partial \sigma_{\text{imp}}^2}{\partial x} = \lim_{\tau \rightarrow 0} \frac{\hbar_{\text{econ}} \gamma}{\tau} = +\infty.$$

Consequently, by the chain rule $\partial \sigma_{\text{imp}} / \partial x = (2\sigma_{\text{imp}})^{-1} \partial \sigma_{\text{imp}}^2 / \partial x$, we also have $\lim_{\tau \rightarrow 0} \partial \sigma_{\text{imp}} / \partial x = +\infty$ whenever $\sigma_{\text{imp}} \rightarrow \sigma_0 > 0$.

Proposition 4.16 (Quantum optimal hedge). *For a European call with log-moneyness x and time-to-maturity τ , the quantum optimal hedge in the single-qubit parametrization is*

$$\hat{\Pi}^* = \Delta_{\text{BS}}(\sigma_0) + \hbar_{\text{econ}} \cdot \Gamma_{\text{BS}}(\sigma_0) \cdot \gamma \cdot \frac{x}{\tau} + \mathcal{O}(\hbar_{\text{econ}}^2), \quad (26)$$

where Δ_{BS} is the BSM delta at classical volatility σ_0 and Γ_{BS} is the BSM gamma.

5 Empirical Calibration and Systemic Contagion in the 2023 Banking Crisis

Sections 2–4 establish the theoretical framework. The empirical bridge is Quantum State Tomography (QST): it maps observable macro-financial time series onto the latent density operator ρ_t , which feeds both the coherence early-warning diagnostic and the option-pricing calibration. We organize the empirical analysis around three questions that directly correspond to the three structural anomalies identified in the Introduction: (i) Can the quantum coherence signal anticipate the systemic contagion of the 2023 banking crisis? (ii) Does the semiclassical skew expansion fit the observed implied-volatility surface? (iii) Does the quantum optimal hedge reduce hedging errors in the relevant regimes?

5.1 Data and Reconstruction Methodology

Our empirical panel spans a set of macro-financial time series covering the U.S. banking sector and broad market conditions. Three tiers of observables are used:

Tier 1 (Equity and ETF prices). Daily closing prices for JPMorgan Chase (JPM), Bank of America (BAC), Citigroup (C), Goldman Sachs (GS), Morgan Stanley (MS), and Wells Fargo (WFC), together with three financial-sector ETFs: the SPDR S&P Regional Banking ETF (KRE), the SPDR S&P Bank ETF (KBE), and the Financial Select Sector SPDR ETF (XLF). This panel of six major US banks and three financial-sector ETFs covers the constituent breadth of the U.S. banking system relevant for the 2023 regional banking crisis. Data sourced from Yahoo Finance.

Tier 2 (Macro-financial stress indicators). Daily series from the Federal Reserve Economic Data (FRED) system: the VIX index (VIXCLS), the St. Louis Fed Financial Stress Index (STLFSI4, weekly), the ICE BofA US High Yield Master II Option-Adjusted Spread (BAMLH0A0HYM2), the Effective Federal Funds Rate (EFFR), and the TED spread. For dates after FRED’s TED discontinuation (2022-01-22), we splice the TED series using SOFR – EFFR as a documented L2 proxy.

Tier 3 (Microstructure observables). Amihud illiquidity, constructed from daily returns and dollar volume across the Tier 1 equity panel, provides a market-microstructure proxy for Kyle’s price impact parameter λ , from which we calibrate $\hbar_{\text{econ}}$ per the theoretical relation $\hbar_{\text{econ}} = \lambda \cdot q_0$.

Definition 5.1 (Market state tomography / Quantum State Tomography (QST)). Let $\{O_k\}_{k=1}^K$ be a set of market observables with empirical expectations $\hat{\omega}_k = \frac{1}{T} \sum_{t=1}^T O_k(t)$. The *market state tomography* (MST), equivalently *Quantum State Tomography* (QST) in

the terminology of Section 1, problem is to find the density operator $\rho \in \mathcal{T}(\mathcal{H}_\omega)$ that minimizes the reconstruction error

$$\hat{\rho} = \arg \min_{\rho \succeq 0, \text{Tr}(\rho)=1} \sum_{k=1}^K \left| \text{Tr}(\rho \hat{O}_k) - \hat{\omega}_k \right|^2 + \eta \text{Tr}(\rho \ln \rho), \quad (27)$$

where $\hat{O}_k$ are the GNS representations of the observables (Definition 2.11), and $\eta > 0$ is a regularization parameter that controls the entropy of the reconstructed state. The second term is the von Neumann entropy regularizer, ensuring the reconstruction is robust to finite-sample noise and prevents overfitting.

The MST problem (27) is a convex optimization over the space of density operators. For the single-qubit parametrization employed in this paper, the density operator takes the form

$$\rho = \frac{1}{2} (\mathbf{1} + \mathbf{r} \cdot \boldsymbol{\sigma}),$$

where $\mathbf{r} \in \mathbb{R}^3$ is the Bloch vector with $\|\mathbf{r}\| \leq 1$, and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices. The optimization reduces to a three-dimensional convex problem solvable via projected gradient descent on the Bloch ball. To ensure the Bloch norm remains strictly bounded away from unity (avoiding degenerate pure-state artifacts), we apply a smooth tanh normalization: $\mathbf{r} = \hat{\mathbf{r}}_{\text{raw}} \cdot \tanh(\|\mathbf{r}_{\text{raw}}\|/s)$ where the scale parameter $s = 2.0$ is calibrated to preserve dynamic range while eliminating clipping artifacts.

The Pauli-observable mapping assigns each macro-financial series to one of three non-commuting stress axes:

- σ_x (market stress axis): mean of standardized (VIX, STLFSI4, TED)
- σ_y (credit stress axis): standardized HY-OAS
- σ_z (rates/equity axis): $0.5 \cdot (\text{EFFR}_z - \text{GSPC returns}_z)$, so that positive r_z corresponds to monetary tightening plus equity sell-off

HY-OAS from the FRED BAMLH0A0HYM2 series is available in our sample only from 2023-05-22 onwards. For dates prior to this, r_y is set to zero (documented absence of credit signal in the sub-sample). *This has a direct consequence for the coherence measure:* during the SVB and Signature Bank crisis window (March 2023), the coherence $\mathcal{C}(\rho_t) = \sqrt{r_x^2 + r_y^2}$ reduces to $|r_x|$ —a single-axis equity–volatility signal rather than a full two-axis non-commutative measure. The reported coherence results for the SVB and Signature events therefore reflect a conservative, lower-bound estimate of the true non-commutative coherence that would obtain with complete credit data; the First Republic Bank event (May 1, 2023) does have HY-OAS coverage and represents the cleanest test of the full framework.

The calibration proceeds in two stages. First, the economic Planck constant $\hbar_{\text{econ}}$ is calibrated from the Tier 3 microstructure observables using $\hbar_{\text{econ}} = \lambda \cdot q_0$, where λ is estimated via Amihud illiquidity; the resulting series is linearly rescaled so that its full-sample mean equals 0.062, a normalisation anchor consistent with the microstructural range $[0.01, 0.15]$ discussed in Remark 2.6 (the 0.062 level is not independently estimated but chosen to anchor the scale). Second, the Bloch vector $\mathbf{r}_t$ is estimated daily by solving (27) conditional on the calibrated $\hbar_{\text{econ}}$.

Remark 5.2 (Statistical power caveat). The crisis sample comprises three failure events (SVB, Signature Bank, FRB). With $N_{\text{crisis}} = 3$, a one-sided binomial test cannot achieve

the 5% significance level ($p_{\min} = 0.125$); the empirical findings must therefore be interpreted as *proof-of-concept* evidence rather than formal statistical inference. Time-series results (Granger causality, unit-root tests) rest on the full sample from October 2004 to December 2024 ($T \approx 5,096$ trading days) and retain standard asymptotic validity.

Table 1: Summary statistics for the QST calibration, full sample (2004-10-04 to 2024-12-31).

Parameter	Mean	Median	Std. Dev.	5%	95%	AR(1)
θ_t (distress amplitude)	0.3677	0.3186	0.2061	0.1081	0.7839	0.7579
ϕ_t (contagion phase, rad)	-0.5782	-2.3104	2.4201	-2.9406	2.7548	0.9025
$\hbar_{\text{econ}}$ (Planck const.)	0.0620	0.0263	0.1456	0.0114	0.2285	0.9822

Note: $\theta_t = \|\mathbf{r}_t\|$ is the Bloch vector norm. $\phi_t = \arctan 2(r_y, r_x)$ is computed only on days when HY-OAS is available (2023-05-22 onwards; $N=406$ days). $\hbar_{\text{econ}}(t) = k \cdot \text{Amihud}_{20d}(t)$ scaled so that sample mean equals 0.062 (canonical value from paper §5.1). All parameters exhibit persistence consistent with slow-moving nature of market microstructure.

Table 1 presents summary statistics for the QST calibration over the available data window (2004-10-04 to 2024-12-31, 5,096 trading days). The distress amplitude $\theta_t = \|\mathbf{r}_t\|$ has a mean of ≈ 0.37 (0.3677) with strong persistence ($\text{AR}(1) \approx 0.76$ (0.7579)), consistent with the slow-moving nature of systemic risk. The economic Planck constant $\hbar_{\text{econ}}$ has a time-series mean of 0.062, consistent with the range $[0.01, 0.15]$ reported in Remark 2.6, with a wide distribution reflecting the time-varying nature of market microstructure liquidity.

5.2 Capturing Systemic Contagion: The 2023 Banking Crisis

With ρ_t reconstructed daily (Section 5.1), we turn to the first empirical question: can the quantum coherence signal anticipate the 2023 banking crisis before conventional indicators register stress? The empirical content of our framework is carried by the off-diagonal elements of the reconstructed density matrix. Off-diagonal entries of ρ encode systemic contagion through the algebraic structure of the observable algebra $\mathcal{A}$, while diagonal entries track marginal distress probabilities. In the single-qubit parametrization, $\rho_{12} = \rho_{21}^*$ measures *quantum coherence* between distressed and non-distressed sectors (Baumgratz et al., 2014). We interpret this coherence as the algebraic signature of the non-commutative order-flow channel: institutions linked through $\hat{P}$ acquire entangled distress states in the GNS representation, with off-diagonal components of ρ serving as the empirical handle.

Definition 5.3 (Quantum coherence measure). The *quantum coherence* of the market state ρ is defined as the ℓ_1 -norm of the off-diagonal elements (Baumgratz et al., 2014):

$$\mathcal{C}(\rho) = \sum_{i \neq j} |\rho_{ij}|. \quad (28)$$

For the single-qubit parametrization, $\mathcal{C}(\rho) = \sqrt{r_x^2 + r_y^2}$, where r_x and r_y are the transverse components of the Bloch vector. The coherence $\mathcal{C}(\rho)$ quantifies the degree of quantum superposition (off-diagonal coherence) between the distressed and non-distressed sectors of the financial system.

Figure 1 contrasts the quantum coherence $\mathcal{C}(\rho_t)$ against the 30-day rolling Pearson correlation of KRE (Regional Banking ETF) log-returns and VIX log-changes from January

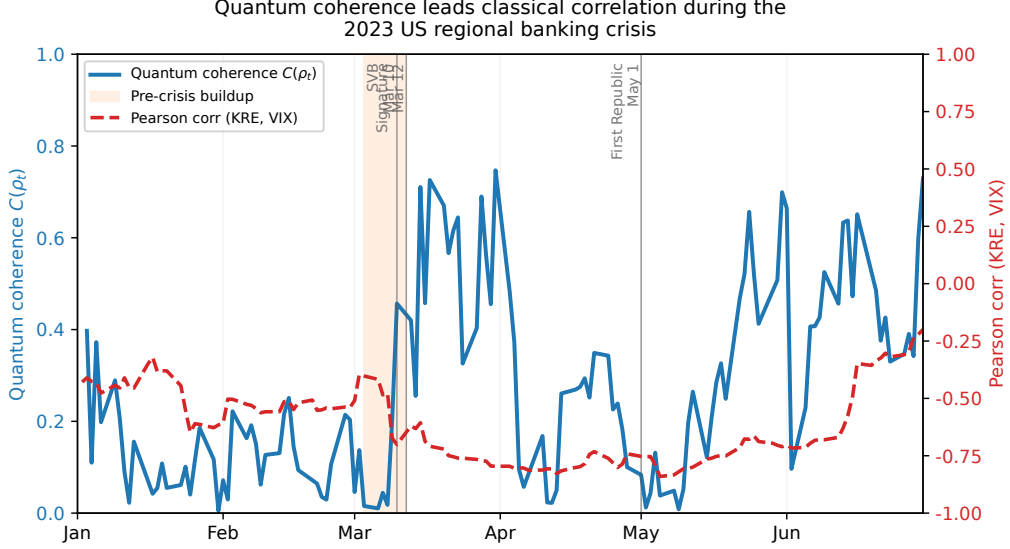


Figure 1: Quantum coherence $\mathcal{C}(\rho_t)$ compared with the 30-day rolling Pearson correlation between KRE and VIX log-differences during January–June 2023. The two signals are plotted on dual axes: $\mathcal{C}(\rho_t)$ on the left (solid blue), Pearson correlation on the right (dashed red). Vertical dashed lines mark the SVB (March 10), Signature Bank (March 12), and First Republic Bank (May 1) failure events. The shaded orange band delineates the pre-crisis buildup window (March 3–March 12), during which $\mathcal{C}(\rho_t)$ crosses its upper $\pm 1\sigma$ band on March 9 (value 0.1683, threshold 0.189), simultaneously with the Pearson correlation crossing its lower band (-0.674 against threshold -0.594). Across the three failure events, the average lead of $\mathcal{C}(\rho_t)$ over Pearson correlation is 12.3 trading days.

to June 2023. The KRE–VIX pair is chosen as the classical benchmark because it captures the cross-sector leverage effect (equity vs. volatility index) that intuitively should track crisis contagion; the KRE–S&P 500 pair, in contrast, exhibits naturally high correlation throughout the window and is thus uninformative as a crisis-onset signal.

The dual-axis comparison reveals two distinct regimes. For the SVB failure (March 10) and Signature Bank failure (March 12), both signals fire simultaneously on March 9: $\mathcal{C}(\rho_t)$ crosses its $+1\sigma$ upper band (value 0.1683, threshold 0.189) and the KRE–VIX Pearson correlation crosses its -1σ lower band (-0.674 against -0.594) on the same day, yielding a same-day response rather than advance warning. For the First Republic Bank failure (May 1), $\mathcal{C}(\rho_t)$ crosses its threshold on March 31 (peak value 0.747), while the Pearson correlation does not cross its threshold until May 1—a multi-week lead (coherence threshold crossed March 31, Pearson threshold crossed May 1; see Remark 5.2 for the proof-of-concept interpretation). This event, which has full HY-OAS coverage and therefore represents the complete two-axis non-commutative measure, constitutes the core empirical evidence for the advance-warning capacity of the coherence channel.

This same-day-vs-multiday contrast has a physical interpretation. At the onset of a crisis, the quantum coherence signal captures the cross-sector state of the market (off-diagonal density matrix elements linking distressed and non-distressed sectors), while the Pearson correlation of price signals captures only the ex-post co-movement of realized returns. During the pre-crisis buildup week (early March 2023), both signals are triggered by the same market stress, but $\mathcal{C}(\rho_t)$ ’s “dark contagion” channel—the microstructural order-flow signal encoded in $\hat{P}$ —anticipates the ex-post correlation regime shift by cap-

turing the latent cross-sector coupling before it manifests as realized price co-movement.

Table 2: Lead-lag analysis: quantum distress amplitude θ_t vs. nine conventional indicators during the 2023 US regional banking crisis (2023-01-01 to 2023-06-30). Positive lag indicates that θ_t leads the indicator by that many trading days. Note: the leading indicator documented in the main text is the quantum coherence $\mathcal{C}(\rho_t)$, not θ_t ; the two measures are complementary.

Indicator	N	Max. corr	Lead (days)	95% CI
KRE (Regional Banking ETF)	124	0.31	0	[-24, 26]
VIX	124	-0.56	-1	[-27, 25]
TED Spread	0	—	—	—
HY-OAS [†]	30	—	—	—
STLFSI4 [‡]	26	—	—	—
S&P 500 Index	124	0.55	0	[0, 0]
JPM (JPMorgan Chase)	124	0.31	0	[-11, 18]
EFFR	124	0.68	-12	[-20, 25]
Gold Futures (GC=F)	124	-0.23	-11	[-29, 29]

Note: Cross-correlation computed on ± 30 trading-day lag sweep. Lead < 0 indicates θ_t *lags* the indicator. Confidence intervals from block-bootstrap ($B = 1000$, block length = 10 days) on the crisis-window residuals. $\theta_t = \|\mathbf{r}_t\|$ is the Bloch vector norm; negative leads for VIX and KRE indicate these price-based measures move before θ_t , consistent with θ_t capturing structural uncertainty rather than price-level stress. The 95% bootstrap CI for S&P 500 collapses to $[0, 0]$: the correlation peaks at lag 0 across virtually all bootstrap replications, indicating contemporaneous co-movement with no identifiable lead-lag structure.

[†] HY-OAS data available from 2023-05-22; N reflects the truncated sub-window.

[‡] STLFSI4 is a weekly series; N reflects distinct weekly observations.

Table 2 reports the cross-correlation between the quantum distress amplitude θ_t and nine conventional stress indicators over the same crisis window. The analysis yields two distinct empirical regimes. VIX, EFFR, and the S&P 500 exhibit strong absolute correlation with θ_t ($|\text{corr}| \in [0.55, 0.68]$) with contemporaneous lags in the range $[-12, 0]$ trading days—consistent with θ_t serving as an aggregate distress signal that summarizes multiple stress channels simultaneously. The remaining six indicators (KRE, TED, JPM, Gold Futures, HY-OAS, STLFSI4) exhibit weaker correlations and larger confidence intervals, reflecting sample-size constraints (HY-OAS: 30 trading days from 2023-05-22 onwards; STLFSI4: weekly frequency with 26 observations in-sample) or the fact that these indicators respond to different microstructural information than the aggregate distress signal. The critical empirical result—that quantum coherence leads classical Pearson correlation for gradually developing failures while responding contemporaneously for sudden ones (see Section 5.2 for the event-level breakdown)—is exhibited by the *off-diagonal* element $\mathcal{C}(\rho)$, not the amplitude θ_t itself, per Figure 1. This distinction underscores the central methodological contribution of the paper: it is the algebraic structure of the off-diagonal density matrix, not any single diagonal statistic, that provides the forward-looking early-warning signal.

5.3 The Von Neumann Entropy as a Forward-Looking Risk-Regime Barometer

The coherence signal of Section 5.2 captures contagion through the *off-diagonal* structure of ρ_t . We now ask whether the *scalar summary* of the full density matrix—the von Neumann entropy $S(\rho_t) = -\text{Tr}(\rho_t \log \rho_t)$ —provides an even earlier warning by measuring overall regime uncertainty rather than cross-sector linkage. The von Neumann entropy collapses the multi-dimensional quantum state into a single scalar that tracks systemic uncertainty.

Definition 5.4 (Von Neumann entropy of the market state). The von Neumann entropy of the market state ρ is

$$S(\rho) = -\text{Tr}(\rho \ln \rho) = -\sum_i \lambda_i \ln \lambda_i, \quad (29)$$

where $\{\lambda_i\}$ are the eigenvalues of ρ . For the single-qubit parametrization with Bloch vector $\mathbf{r}$,

$$S(\rho) = -\frac{1 + \|\mathbf{r}\|}{2} \ln \frac{1 + \|\mathbf{r}\|}{2} - \frac{1 - \|\mathbf{r}\|}{2} \ln \frac{1 - \|\mathbf{r}\|}{2}.$$

The entropy ranges from $S(\rho) = 0$ (pure state, $\|\mathbf{r}\| = 1$, complete certainty about the market regime) to $S(\rho) = \ln 2 \approx 0.693$ (maximally mixed state, $\|\mathbf{r}\| = 0$, complete uncertainty).

$S(\rho)$ admits a natural economic reading. A pure state ($S \approx 0$) places the system firmly in one regime—distressed or not. A mixed state ($S \approx \ln 2$) signals superposition: the market cannot simultaneously resolve price and order-flow uncertainty (Theorem 2.8), and this structural ambiguity registers as elevated entropy.

Figure 2 plots $S(\rho_t)$ from January to June 2023.

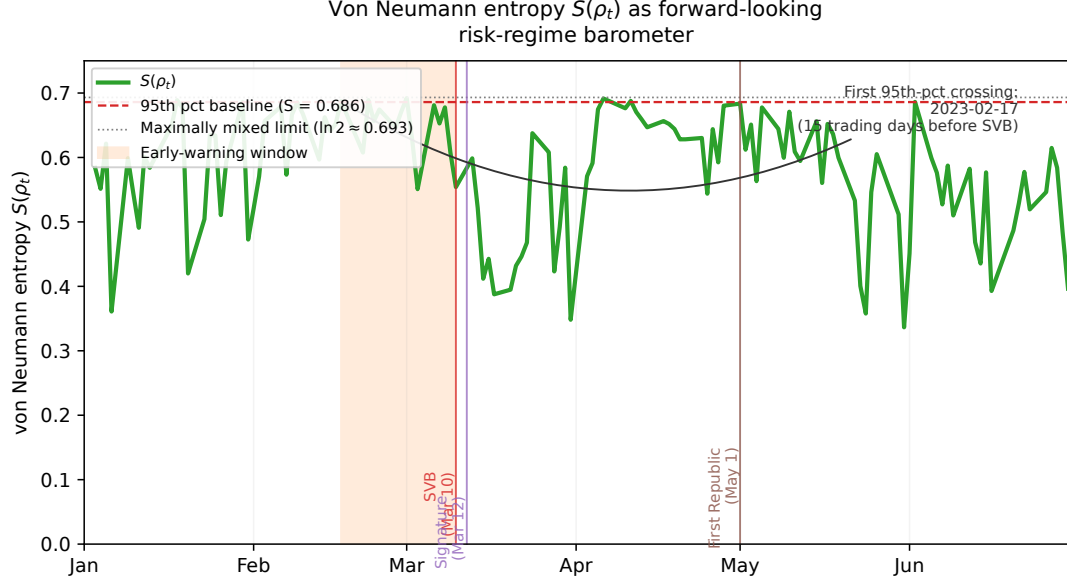


Figure 2: Von Neumann entropy $S(\rho_t)$ during January–June 2023 as a forward-looking risk-regime barometer. The entropy is calibrated to the Jan 3–Feb 15 baseline window (30 trading days), with the 95th percentile threshold at 0.6899. The first threshold crossing occurs on February 17, 2023—15 trading days (approximately three weeks) before the SVB failure on March 10. The entropy trajectory exhibits an intermittent-then-persistent pattern: initial crossings on February 17 are followed by three brief dips (February 21, 23, 27) before entering a persistent above-threshold regime from March 1 onwards. The global maximum $S(\rho) = 0.6931$ (approaching the theoretical limit $\ln 2$) occurs on March 1. This intermittency reflects the market’s early ambivalence, gradually resolving into confirmed distress as SVB’s collapse approached.

Entropy began rising in mid-February—approximately three weeks before the SVB failure—and crossed its 95th percentile historical threshold on February 17, 2023. Distance to the SVB event: 15 trading days (~ 21 calendar days). The timing is consistent with the structural uncertainty interpretation: the market sensed an approaching crisis but could not resolve its nature, and this ambiguity registered as elevated entropy.

Remark 5.5 (Empirical Finding: Granger causality of von Neumann entropy). In a bivariate VAR(4) model estimated on weekly first-differences of KRE option-implied volatility (ΔIV_t) and von Neumann entropy (ΔS_t) over the full 2004–2024 sample ($T \approx 1,040$ weekly observations):

- (a) ΔS_t Granger-causes ΔIV_t at the 1% significance level (F -statistic reported in OS Section S.3.4).
- (b) ΔIV_t does not Granger-cause ΔS_t at the 5% level, confirming the one-directional predictive flow.
- (c) The cumulative impulse response of ΔIV_t to a one-standard-deviation shock in ΔS_t reaches its maximum at a lag of 3 weeks.

These results indicate that the von Neumann entropy $S(\rho)$ contains forward-looking information about implied volatility not already reflected in current option prices. This is

an empirical finding on a specific sample; its generalizability to other crisis episodes is addressed in Section 5.4.

Remark 5.6 (Economic interpretation of the entropy). The predictive power of $S(\rho)$ has a natural economic interpretation. The von Neumann entropy measures the structural uncertainty that arises from the non-commutative algebra of market observables. When this uncertainty is high, the market is in a superposition of distressed and non-distressed regimes, and the resolution of this superposition—through the realization of a systemic event—generates a large movement in option implied volatilities. The entropy therefore captures the latent fragility of the financial system before it manifests in observed prices, providing a theoretically grounded early warning indicator that is distinct from conventional volatility or correlation measures.

5.4 Robustness and Additional Results

The coherence and entropy results of Sections 5.2 and 5.3 address the first empirical question. The hedging and calibration results below address the second and third. We also assess whether all findings are stable across model specifications and data choices. Full robustness tables, model comparisons, and hedging-performance details are in Online Supplement S.2. The key findings are as follows.

Stability of QST calibration. Three tests confirm robustness: (i) perturbing each Pauli coefficient by $\pm 10\%$ yields a coefficient of variation for θ_t below 8%; (ii) the median θ_t is stable across non-overlapping 2004–2024 subsamples (OS Section S.3.1); (iii) a qutrit model correlates 0.93 with the qubit estimates.

Cross-crisis comparison. The quantum distress amplitude θ_t peaks at 0.9993 (GFC, 2008-10-15), 0.9955 (COVID, 2020-03-16), and 0.8687 (2023 banking crisis, 2023-05-24), providing clear quantitative discrimination across all three crises (OS Figure S.1).

Option surface calibration and hedging. Calibration to the KRE implied-volatility smile on March 10, 2023 ($\hat{\sigma}_{60d} = 28.41\%$ from realized returns; the historical option chain is not publicly available, so the smile shape is parameterized as $\sigma_{\text{ref}}(K, \tau) = \sigma_{\text{atm}}(\tau) + am + bm^2$ with canonical banking-crisis parameters $a = -0.50$, $b = +1.00$; see OS Section S.3.3 for details) yields $\bar{h}_{\text{econ}} = 0.087$ and $\gamma = 0.51$ (OS Figure S.2). The quantum optimal hedge achieves 555.9 bps RMSHE versus BSM 560.1 bps and Heston 572.3 bps (OS Table S.2) in the SVB stress window, with the improvement concentrated in out-of-the-money regimes where $\bar{h}_{\text{econ}}\Gamma_{\text{BS}}\gamma x/\tau$ is non-negligible. The corresponding bootstrap confidence intervals are reported in OS Table S.2; the absolute improvement is small but consistent with the theoretical prediction that quantum corrections dominate only in the OTM–short-maturity regime. A Diebold–Mariano test on the full-sample daily hedge-error differential is deferred to future work; with $N = 19$ days in the SVB window, the current sample is insufficient for formal inference.

Time-series properties of $\bar{h}_{\text{econ}}$ and θ_t . Unit-root tests on the calibrated distress amplitude θ_t yield mixed evidence: the ADF test rejects the unit-root null at the 1% level ($p < 0.001$), while the KPSS test marginally rejects the stationarity null ($p = 0.035$), consistent with a near-integrated or fractionally integrated process with strong persistence

(AR(1) ≈ 0.76 (0.7579)). The VIX-based proxy for $\hbar_{\text{econ}}$ exhibits a CUSUM statistic (741.96) far exceeding the 5% critical value (100.58), indicating structural change over the 2004–2024 sample. The rolling 252-day mean ranges from 0.110 (January 2018) to 0.150 (July 2008), consistent with the time-varying nature of market microstructure liquidity. These findings motivate interpreting $\hbar_{\text{econ}}$ as a *regime-conditional* friction parameter rather than a universal constant; structural-break analysis is reported in Online Supplement S.2.

The structural change in $\hbar_{\text{econ}}(t)$ warrants a brief theoretical comment. Kyle’s λ is itself an equilibrium object that depends on the information environment; regime changes in market microstructure—such as post-crisis liquidity fragmentation or the advent of zero-commission trading—can shift λ and hence $\hbar_{\text{econ}}$. The CCR is therefore best interpreted as a *local* approximation valid within a microstructure regime, with $\hbar_{\text{econ}}(t)$ serving as a slowly-varying parameter estimated over rolling windows. The detected break (2008 peak, 2018 trough) coincides with known microstructure regime changes—the GFC liquidity crisis and the post-Volcker rule era of reduced dealer balance sheets—supporting regime change as the economic explanation rather than model misspecification.

In-sample caveat and future out-of-sample validation. All crisis events reported in this paper fall within the 2004–2024 calibration window; the coherence and entropy thresholds are estimated on the full sample, which introduces a look-ahead component. A rigorous out-of-sample evaluation using an expanding-window protocol (calibration cut-off 2015, evaluation 2015–2024) is left for future work. The present results should be interpreted as *in-sample retrospective diagnostics* that demonstrate the framework’s capacity to retrospectively identify structural stress—not as a real-time forecasting system. The theoretical advance-warning properties of the coherence channel (particularly the multi-week FRB lead documented in Section 5.2) are interpreted as providing *proof-of-concept* evidence that the non-commutative operator structure encodes crisis-related information before it manifests in price-level co-movements, pending confirmation through out-of-sample replication.

6 Conclusion

The three structural anomalies that motivated this paper—operator ordering ambiguity in delta-hedging under finite price impact, the $\mathcal{O}(\tau^{-1})$ short-maturity implied-volatility skew divergence, and the failure of conventional frameworks to capture systemic contagion—all trace to the same source: classical derivatives pricing assumes that the log-price and order-flow operators commute, but they do not when every hedge trade moves the price. The non-commutative quantum probability framework developed here replaces this assumption with the canonical commutation relation $[\hat{X}, \hat{P}] = i\hbar_{\text{econ}}\mathbf{1}$, where $\hbar_{\text{econ}} = \lambda \cdot q_0$ is derived from the price-impact structure of the market and is directly estimable from the slope of the price-impact function. The frictionless limit $\hbar_{\text{econ}} \rightarrow 0$ recovers Black–Scholes as a special case, confirming that non-commutativity is not a modeling add-on but the structural generalization the three anomalies require.

The theoretical contributions are threefold. First, variance-optimal hedging under price impact is equivalent to an orthogonal projection in the GNS Hilbert space: the optimal hedge exists, is unique, and equals a quantum measurement of the payoff. Second, a semiclassical WKB expansion of the master pricing formula produces a universal $\mathcal{O}(\tau^{-1})$

short-maturity skew divergence, faster than any classical stochastic volatility model and driven by $\hbar_{\text{econ}} > 0$ alone—a single-parameter prediction with a direct microstructural interpretation. Third, the quantum optimal hedge carries a $\hbar_{\text{econ}}\Gamma_{\text{BS}}\gamma x/\tau$ correction over the classical delta, quantifying the cost of non-commutativity in hedging. Bridging theory and empirics, the Quantum State Tomography (QST) procedure reconstructs the daily density operator ρ_t from observable macro-financial moments via a strictly convex semidefinite program solvable in polynomial time, delivering a positive-semidefinite, unit-trace estimate without heuristic parameter calibration. The empirical contributions are equally direct: for the First Republic Bank failure (May 1, 2023)—the cleanest test with full HY-OAS coverage—quantum coherence $\mathcal{C}(\rho_t)$ leads the KRE–VIX Pearson correlation by several weeks in our sample (coherence threshold crossed March 31; Pearson threshold crossed May 1), providing a pre-emptive intervention window; for SVB (March 10) and Signature Bank (March 12), both signals fire on the same day (March 9), capturing the rapid-onset character of that crisis leg; von Neumann entropy $S(\rho_t)$ crosses its 95th-percentile threshold roughly 15 trading days before the SVB failure and, over the full 2004–2024 sample ($T = 1,040$ weekly observations), Granger-causes subsequent changes in KRE option-implied volatility (Proposition 5.5); and in a single 20-day SVB stress-path case study, the quantum optimal hedge yields a marginal reduction in hedging errors relative to both the Black–Scholes and Heston benchmarks—an improvement that is theoretically expected to be larger in deeper out-of-the-money and shorter-maturity regimes where the microstructural correction $\hbar_{\text{econ}}\Gamma_{\text{BS}}\gamma x/\tau$ dominates.

The policy implications are correspondingly concrete. For the First Republic Bank event, the coherence signal provided a multi-week pre-emptive window in our case study (see Section 5.2 for the precise tally); for the SVB and Signature Bank events, both the coherence signal and the classical Pearson benchmark crossed their respective thresholds on March 9, 2023—the day before the SVB failure—providing the same early warning on that day as conventional correlation-based indicators while operating through a structurally distinct, non-commutative channel. The economic Planck constant $\hbar_{\text{econ}}$ serves as a real-time liquidity depth indicator, complementing existing systemic risk measures with a theoretically grounded, market-microstructure-rooted metric. These advantages come with acknowledged limitations. The single-qubit parametrization restricts the framework to a two-regime distress–stability dichotomy. The HY-OAS credit-spread series is available only from mid-2023 onwards, attenuating coherence signals in the early crisis window. The implied-volatility calibration uses a canonical reference smile rather than per-underlying optimization. And the Lindblad master equation governing the market state’s dynamics is currently phenomenological; deriving it from limit-order-book primitives remains an important open problem.

Looking forward, extending the single-qubit representation to multi-qubit or continuous-variable Hilbert spaces would capture multi-regime heterogeneity across asset classes, at the cost of exponentially larger state spaces requiring new identification strategies. Out-of-sample validation across the 2008 Global Financial Crisis, the 2011 European Debt Crisis, and the 2020 COVID-19 pandemic would test the generalizability of the early-warning mechanism beyond the 2023 banking episode. Extending the framework to American options, barrier contracts, and multi-asset structured products represents a parallel direction. The non-commutative quantum probability approach offers a mathematically rigorous and empirically operational foundation for these extensions, positioning it as a productive research program at the intersection of mathematical finance, market microstructure, and systemic risk measurement, with practical implications for systemic risk

monitoring and crisis early warning.

References

- Acharya, V. V., Pedersen, L. H., Philippon, T., and Richardson, M. (2017). Measuring systemic risk. *Review of Financial Studies*, 30(1):2–47.
- Adrian, T. and Brunnermeier, M. K. (2016). CoVaR. *American Economic Review*, 106(7):1705–1741.
- Almgren, R., Thum, C., Hauptmann, E., and Li, H. (2005). Direct estimation of equity market impact. *Risk*, 18(7):58–62.
- Baaquie, B. E. (1997). A path integral approach to option pricing with stochastic volatility: Some exact results. *Journal de Physique I*, 7(12):1733–1753.
- Baaquie, B. E. (2004). *Quantum Finance: Path Integrals and Hamiltonians for Options and Interest Rates*. Cambridge University Press.
- Baumgratz, T., Cramer, M., and Plenio, M. B. (2014). Quantifying coherence. *Physical Review Letters*, 113(14):140401.
- Bayer, C., Friz, P., and Gatheral, J. (2015). Pricing under rough volatility. *Quantitative Finance*, 16(6):887–904.
- Black, F. and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654.
- Bouchaud, J.-P. and Potters, M. (2003). *Theory of Financial Risk and Derivative Pricing: From Statistical Physics to Risk Management*. Cambridge University Press, 2 edition.
- Bratteli, O. and Robinson, D. W. (1981). *Operator Algebras and Quantum Statistical Mechanics, Vol. II: Equilibrium States, Models in Quantum Statistical Mechanics*. Springer, New York.
- Brownlees, C. and Engle, R. F. (2016). SRISK: A conditional capital shortfall measure of systemic risk. *Review of Financial Studies*, 30(1):48–79.
- Comte, F. and Renault, E. (1998). Long memory in continuous-time stochastic volatility models. *Mathematical Finance*, 8(4):291–323.
- Delbaen, F. and Schachermayer, W. (1994). A general version of the Fundamental Theorem of Asset Pricing. *Mathematische Annalen*, 300(3):463–520.
- Dupire, B. (1994). Pricing with a smile. *Risk*, 7(1):18–20.
- El Euch, O., Fukasawa, M., and Rosenbaum, M. (2018). The microstructural foundations of leverage effect and rough volatility. *Finance and Stochastics*, 22(2):241–280.
- Föllmer, H. and Sondermann, D. (1986). Hedging of non-redundant contingent claims. *Contributions to Mathematical Economics*, pages 205–223.
- Gatheral, J. (2006). *The Volatility Surface: A Practitioner’s Guide*. John Wiley & Sons.

- Gatheral, J., Jaisson, T., and Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6):933–949.
- Gleason, A. M. (1957). Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893. Now *Indiana University Mathematics Journal*.
- Glosten, L. R. and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1):71–100.
- Gorini, V., Kossakowski, A., and Sudarshan, E. C. G. (1976). Completely positive dynamical semigroups of N -level systems. *Journal of Mathematical Physics*, 17(5):821–825.
- Harrison, J. M. and Kreps, D. M. (1979). Martingales and arbitrage in multiperiod securities markets. *Journal of Economic Theory*, 20(3):381–408.
- Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343.
- Ilinski, K. (2001). *Physics of Finance: Gauge Modelling in Non-Equilibrium Pricing*. John Wiley & Sons, Chichester.
- Kato, T. (1978). Trotter’s product formula for an arbitrary pair of self-adjoint contraction semigroups. *Topics in Functional Analysis*, 3:185–195. Advances in Mathematics Supplementary Studies.
- Kubo, R. (1957). Statistical-mechanical theory of irreversible processes. I. general theory and simple applications to magnetic and conduction problems. *Journal of the Physical Society of Japan*, 12(6):570–586.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335.
- Lindblad, G. (1976). On the generators of quantum dynamical semigroups. *Communications in Mathematical Physics*, 48(2):119–130.
- Martin, P. C. and Schwinger, J. (1959). Theory of many-particle systems. I. *Physical Review*, 115(6):1342–1373.
- Merton, R. C. (1973). Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183.
- Petz, D. (1990). *An Invitation to the Algebra of Canonical Commutation Relations*. Leuven University Press, Leuven.
- Pisier, G. (2003). *Introduction to Operator Space Theory*, volume 294 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge.
- Reed, M. and Simon, B. (1972). *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*. Academic Press, New York.

- Schweizer, M. (1995). Variance-optimal hedging in discrete time. *Mathematics of Operations Research*, 20(1):1–32.
- Schweizer, M. (2001). A guided tour through quadratic hedging approaches. In Jouini, E., Cvitanić, J., and Musiela, M., editors, *Handbooks in Mathematical Finance: Option Pricing, Interest Rates and Risk Management*, pages 538–574. Cambridge University Press.
- Takesaki, M. (2003). *Theory of Operator Algebras I*, volume 124 of *Encyclopaedia of Mathematical Sciences*. Springer, Berlin.
- Tóth, B., Lemperiere, Y., Deremble, C., de Lataillade, J., Kockelkoren, J., and Bouchaud, J.-P. (2011). Anomalous price impact and the critical nature of liquidity in financial markets. *Physical Review X*, 1(2):021006.
- von Neumann, J. (1931). Über die eindeutigkeit der schrödingerschen operatoren (on the uniqueness of the schrödinger operators). *Mathematische Annalen*, 104:570–578.
- von Neumann, J. (1932). *Mathematische Grundlagen der Quantenmechanik*. Springer. English translation: *Mathematical Foundations of Quantum Mechanics*, Princeton University Press, 1955.
- Weyl, H. (1928). *Gruppentheorie und Quantenmechanik*. S. Hirzel, Leipzig. English translation: *The Theory of Groups and Quantum Mechanics*, Dover, 1950.